RIDOT AI SYMPOSIUM · PEER EXCHANGE

PROCEEDINGS

Intelligent Mobility: AI Meets Real Transportation

Rhode Island Department of Transportation · Artificial Intelligence Symposium and Peer Exchange

DATES

August 19–20, 2026

Tuesday & Wednesday · two full days

LOCATION

New England Institute of Technology

West Greenwich, Rhode Island

FORMAT

Keynotes · Panels · Peer exchange

14 sessions over two days

ORGANIZING TEAM

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PROGRAM & MATERIALS AT

www.ridotaisymposium.org

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Introduction

The Rhode Island Department of Transportation (RIDOT) convened a two-day symposium and peer exchange at the New England Institute of Technology on August 19–20, 2026, bringing together leaders and technical experts from federal, state and international transportation agencies; universities; non-profit transportation associations and non-governmental organizations; and technology partners to examine where artificial intelligence meets the real work of building, running, and maintaining transportation systems.

Each day opened with a keynote and fireside chat and closed with a reflection on key takeaways. In between, technical sessions covered AI applications for proactive safety, ITS and mobility operations, planning and project development, transportation asset management, digital twins, and data analytics. The discussion was consistently practical: what problem is being solved, what evidence exists that AI solved it, and what it took organizationally to get there.

Six themes ran through the two days. Presenters urged agencies to start with the “why” rather than the technology; recognized that new computing and AI capability is making previously undiscoverable information usable; stressed that value from AI depends on a governed data foundation; agreed that organizational and workforce change is the larger share of the work; noted that AI capability is outpacing governance, procurement, and institutional adoption; and called for far better mechanisms to share tools, pilots, and lessons across agencies.

These proceedings document the symposium sessions in the order they occurred. Each session opens with a description, followed by a summary of key takeaways and then a record of individual presentations and the discussion that followed. A roster of speakers and a list of the AI initiatives, policies, and tools referenced during the event appear at the front of the document.

2	14	43	6
DAYS	SESSIONS	SPEAKERS	THEMES

> The technology part of AI deployment is only 20% of the effort; the remaining 80% involves organizational and process change and workforce transformation.
>
> CONFERENCE THEME 4 · PREPARE THE ORGANIZATION AND THE WORKFORCE

Speakers

Speakers and panelists are listed by sector, with the role and organization noted as of the symposium.

USDOT	
Sophie Abo	Transportation Planner · USDOT Volpe National Transportation Systems Center (VOLPE)
Matthew Cuddy	Innovation Management and AI Applications Strategist · VOLPE
Daniel (Mac) Lang	Policy Analyst · VOLPE
Sean McMaster	Administrator · United States Department of Transportation (USDOT) Federal Highway Administration (FHWA)
Seval Oz	Assistant Secretary for Research & Technology · USDOT
Lisa Randall	AI Program Manager · USDOT Intelligent Transportation Systems Joint Program Office (ITS JPO)
Robert Sheehan	Director, AI Deployment Center of Excellence (AI COE) · FHWA
Craig Thor	Chief Scientist · FHWA Turner-Fairbank Highway Research Center (TFHRC)
Derek Torrey	Rhode Island Division Administrator · FHWA

US STATE AGENCIES	
Justin Bruner	Asset Management Engineer · Pennsylvania Department of Transportation (PennDOT)
Gregory Ciparelli	Transportation Planning Director and Chief Data Officer · Connecticut Department of Transportation (CTDOT)
Pamela Cotter	Administrator of Planning · Rhode Island Department of Transportation (RIDOT)
Scott Gausland	Chief Data and Analytics Officer · Rhode Island Department of Administration
Charles Major	Manager of GIS Data Engineering · Massachusetts Department of Transportation (MassDOT)
Robert Rocchio	Interim Director · RIDOT
Jocelyn Stein	Business Integration & Digital Accessibility Lead · Minnesota Department of Transportation (MnDOT)
Giri Venkiteela	Innovation Officer · New Jersey Department of Transportation (NJDOT) Bureau of Research and Innovation
Jennifer Volkening	Director of Data Analytics and Governance · Utah Department of Transportation (UDOT)
Christos Xenophontos	Assistant Director, Administrative Services · RIDOT

ASSOCIATIONS / NON-GOVERNMENTAL AGENCIES	
Laura Chace	President and CEO · Intelligent Transportation Society of America (ITSA)
Lawrence Dwyer	Director of Safety and Mobility · American Association of State Highway and Transportation Officials (AASHTO)
Patricia Hendren	Executive Director · The Eastern Transportation Coalition (TETC)
Katherine Kortum	Senior Program Officer · Transportation Research Board (TRB)

INTERNATIONAL	
Thomas Gruenstaeudl	Head of Project Development · ASFINAG GmbH (Austria)
Alexander Walcher	Managing Director · ASFINAG GmbH (Austria)

ACADEMIA	
Mehrshad Amini	Assistant Professor · University of Rhode Island (URI)
Christopher Baxter	Professor · University of Rhode Island (URI)
Timothy Henry	Professor and AI Program Director · Rhode Island College (RIC), academic team lead for the Rhode Island Governor's AI Task Force
Henry Young	Assistant Provost · New England Institute of Technology (NEIT)
Jinhua Zhao	Professor of Cities and Transportation · Massachusetts Institute of Technology (MIT)

PRIVATE SECTOR	
Matthew Deal	Enterprise Government Account Manager - New England States · Esri
Donovan Guin	US Public Transportation Lead · IBM
Arnulf Hagen	Founder and Chief Executive Officer · Aidoalos (Norway)
Scott Hensley	Associated Partner · IBM
Justin O'Daniel	Enterprise Account Executive · Databricks
Hyun-A Park	President · Spy Pond Partners, LLC
Grace Sammartino	Senior Account Executive · Environmental Systems Research Institute (Esri)
Mike Snodgrass	AI Specialist · Google

Conference Overview

Sessions

The Rhode Island Department of Transportation (RIDOT) convened a two-day symposium and peer-exchange involving USDOT and state DOT leadership, engineers, planners, and researchers to discuss the future of artificial intelligence (AI) for transportation - where AI meets the real work of building, running, and maintaining transportation systems.

The conference was hosted by the New England Institute of Technology (NEIT) campus in West Greenwich, RI. Over 100 individuals were in attendance.

Each day of the symposium began with a distinguished keynote speaker and fireside chat and ended with a wrap-up session to reflect on key takeaways. In between, there were six technical sessions covering AI applications for proactive safety, ITS and mobility, planning and project development, transportation asset management, digital twins, and data analytics. The following tables provide a summary of key conference themes and AI initiatives of note.

Conference Themes

Presentations and discussions at the symposium can be grouped into six major themes:

01 **Focus on the "WHY" for AI.**
Participants cautioned against adopting new technologies for their own sake They stressed focusing on real business problems and making sure AI is the right solution. They also discussed the importance of demonstrating ROI from AI pilots before scaling applications, and learning how to communicate about the value to leadership.

02 **Leverage New Capabilities that AI Enables.**
Advancement in computing power together with AI techniques is opening up new opportunities to make discoverable what was previously undiscoverable. This includes information from imagery, large data streams, and documents. AI techniques are enabling agencies to become more proactive rather than reactive. Use of Large Language Models (LLMs) is rapidly expanding – allowing people to ask questions and receive answers in natural language.

03 **Solidify the Data Foundation.**
Getting value from AI depends on well-organized and managed data. This has elevated the importance of efforts to integrate, standardize, and govern data resources. Such efforts are not only underway within transportation agencies; there are national efforts to improve data interoperability both within transportation and across sectors, and to build integrated data platforms to enable advanced analytics and greater connectivity across state borders.

04 **Prepare the Organization and the Workforce.**

There was consensus that the technology part of AI deployment is only 20% of the effort; the remaining 80% involves organizational and process change and workforce transformation. Many transportation agencies are working to build AI literacy and technical skills and universities are playing important roles to develop the future workforce with these capabilities. Part of preparing organizations for AI deployment is an emphasis on keeping "humans in the loop – or at the helm," teaching staff to critically evaluate model results, communicating that AI is intended to augment rather than replace existing staff, and adopting a cautious and transparent approach to AI implementation that recognizes and addresses real concerns.

05 **The Rapid Pace of AI Development.**

There was recognition that AI capability is advancing faster than governance, procurement, alignment, and institutional adoption. Participants talked about not trying to "boil the ocean" but emphasize small scale pilots that can solve real problems, put some guardrails in place to manage risk, build the skills and literacy needed to enable experimentation, and maintain the agility to change course as things evolve.

06 **Importance of Sharing Information and Tools.**

Given the rapid evolution of AI capabilities and the large number of research studies and pilots, it is essential to maintain frequent collaboration and communication across organizations and provide a way to transfer best practices and tools. USDOT provided information about several important initiatives to facilitate this kind of collaboration, but there are also important roles for other organizations to play.

AI Initiatives and Systems of Note

The following are initiatives or systems that were referenced in the symposium sessions

FEDERAL

- **The Genesis Mission**
  https://www.energy.gov/undersecretaryforscience/genesis-mission/genesis-mission
- **USDOT's Artificial Intelligence Policy**
  https://www.transportation.gov/orders/ai-policy
- **FHWA's Artificial Intelligence Deployment Center of Excellence**
  https://highways.dot.gov/ai-coe
- **National AI Transportation Coalition Pooled Fund Study**
  https://pooledfund.org/Details/Solicitation/1662
- **State DOT Council for Strategic AI Adoption Pooled Fund Study**
  https://pooledfund.org/Details/Study/1806
- **Advancement of Tools for Transportation Asset Management Pooled Fund Study**
  https://pooledfund.org/Details/Study/1801

- **USDOT JPO AI for Transportation Planning and Design Initiative: see: https://www.its.dot.gov/research-areas/artificial-intelligence/focus-areas/put-ai-into-practice/transportation-planning-design/**
  https://www.its.dot.gov/research-areas/artificial-intelligence/focus-areas/put-ai-into-practice/transportation-planning-design/
- **USDOT JPO AI Deployment Strategies and Projects**
  https://its-dr.fhwa.dot.gov/research-areas/artificial-intelligence/focus-areas/put-ai-into-practice/
- **USDOT JPO AI Readiness and Assessment Tools**
  https://its-dr.fhwa.dot.gov/research-areas/artificial-intelligence/focus-areas/learn-connect/

STATE

- **Florida DOT AI Use Case Collection**
  https://fdotwww.blob.core.windows.net/sitefinity/docs/default-source/design/training/transportationsymposium/2025-june/19-08-dilmore-ai-current-potential-future-use-cases.pdf
- **Michigan DOT - Enhance Bridge Image Attribution Through Automated Post Image Processing**
  https://rosap.ntl.bts.gov/view/dot/91813
- **AI the MN Way Roadmap**
  https://mn.gov/mmb/planning/projects/ai-roadmap/
- **Minnesota DOT GenAI Standard**
  https://www.dot.state.mn.us/policy/it-data/it-003.html
- **Missouri DOT I-270 Predictive Layered Operations Initiative (PLOI)**
  https://www.itskrs.its.dot.gov/2026-b02046
- **New Jersey Innovation Skills and Accelerator Training**
  https://innovation.nj.gov/skills/ai/
- **Rhode Island Data Analytics and AI Center of Excellence**
  https://etss.ri.gov/data-AI
- **Texas DOT AI Strategic Plan**
  https://www.txdot.gov/about/leadership/strategic-plans/txdot-artificial-intelligence-strategic-plan-update.html
- **Utah DOT AI Strategy and Roadmap**
  https://drive.google.com/file/d/1t_FDIhMDobwXPgfmRoj2sQH7oynOgWXa/view

ASSOCIATIONS

- **ITS America AI Implementation Guide**
  https://itsa.org/advocacy-material/a-guide-to-practical-next-steps-for-ai-implementation/
- **ITS America AI Maturity Self-Assessment Tool**
  https://itsa.org/advocacy-material/path-to-ai-maturity-self-assessment-worksheet/

PRIVATE

- **Aidalos system (multi-agent AI system demonstrated for RIDOT bridges)**
  https://www.aidalos.com/
- **CollisionEye Intelligent Collision Detection**
  https://www.eyyes.com/en/products/collisioneye/

INTERNATIONAL

- **ASFINAG (Austria) knowledge assistants Agnes and Adam**
  https://www.youtube.com/watch?v=NBh3e5VmR14

RIDOT AI SYMPOSIUM & PEER EXCHANGE

Day 1

Tuesday, August 19, 2026

01 Opening Session
02 Day 1 Keynote and Fireside Chat — FHWA Administrator Sean McMaster
03 Governance Frameworks for Successful AI Development and Deployment in Transportation
04 From Reactive to Predictive: Using AI to Advance Transportation Safety
05 Optimizing ITS and Mobility Systems with AI
06 Making AI Real: From Concepts to Practical Transportation Applications
07 Building an AI Strategy: Governance, Workforce, Implementation, and Organizational Readiness
08 Day 1 Reflections

DAY 1 · SESSION 01

Greetings and Welcoming Remarks

Opening Session

SESSION DESCRIPTION

The opening session welcomed participants to the RIDOT Symposium/Peer Exchange on Artificial Intelligence in the Road & Mobility Sector and established the context for the two-day event. Senior leaders from the Federal Highway Administration (FHWA), New England Institute of Technology (NEIT), and Rhode Island Department of Transportation (RIDOT) talked about the potential for artificial intelligence to change how transportation agencies plan, design, construct, operate, and maintain transportation systems.

Speakers emphasized the importance of collaboration among federal and state transportation agencies, researchers, universities, technology companies, and other partners in moving AI applications from research and experimentation into practical use. The opening remarks framed the symposium as an opportunity not simply to learn about emerging technology, but to identify practical applications and lessons that participants could take back to their organizations.

SESSION TAKEAWAYS

- **AI has the potential to affect virtually every aspect of transportation agency operations.** Applications discussed included predictive maintenance, traffic management, advanced analytics, infrastructure monitoring, and improved planning and operational decision-making.
- **The emphasis should be on practical value rather than technology for its own sake.** RIDOT described a progression from research to pilots and ultimately to incorporation of

successful technologies into routine agency operations.

- **AI can help transportation agencies make better use of increasingly large amounts of data.** Agencies now possess extensive information about bridges, pavements, traffic, and system users; AI provides new ways to turn those data into timely and actionable information.
- **Workforce constraints are an important reason for transportation agencies to explore AI.** AI can potentially automate repetitive or lower-value activities, allowing scarce technical staff to devote more attention to work requiring professional judgment, expertise, and creativity.
- **AI adoption needs to be responsible and deliberate.** Speakers acknowledged that AI creates legitimate questions as well as opportunities. Transportation agencies need to understand the technology, evaluate where it adds value, and determine how it can be used responsibly.
- **Peer exchanges are particularly important for emerging technologies.** FHWA views events such as the symposium as an important form of technical assistance that allows agencies to learn from one another and accelerate adoption of useful innovations.
- **Federal-state partnerships can help move innovation into practice.** FHWA's role extends beyond providing funding to technical assistance, research, expertise, and knowledge transfer that can help states implement new approaches.

Derek Torrey — Division Administrator, FHWA Rhode Island Division

Derek Torrey welcomed participants on behalf of FHWA and described the broader federal-state partnership within which transportation innovation occurs. He explained that the Federal-aid Highway Program is federally assisted but state administered: state DOTs own and operate infrastructure, plan and construct projects, and maintain roads and bridges, while FHWA helps states accomplish that work.

Torrey emphasized that FHWA's contribution goes well beyond funding. Technical assistance is an equally important part of the agency's role. FHWA division staff provide planning, permitting, engineering, and other assistance, while national experts can be brought in to address specialized transportation issues.

He highlighted the role of the Turner-Fairbank Highway Research Center in identifying and developing innovations that can help states deliver transportation projects more quickly, efficiently, and cost-effectively. A key challenge is translating research and innovation into applications that can actually be deployed on transportation projects.

Torrey characterized peer exchanges such as the RIDOT symposium as another important form of FHWA technical assistance. AI is a particularly significant emerging issue because of its potential to change how transportation projects are delivered nationwide. He encouraged attendees to participate actively and, most importantly, leave with practical ideas that they could bring back to their organizations.

Dr. Henry Young — Assistant Provost, New England Institute of Technology

Dr. Henry Young welcomed participants on behalf of New England Institute of Technology and described transportation as being at a pivotal point as infrastructure becomes increasingly connected, data-driven, and intelligent.

He characterized AI as a technology that is already transforming transportation planning, design, construction, operations, and maintenance. He cited predictive maintenance, intelligent traffic management, advanced analytics, and emerging mobility solutions as examples of areas in which AI can improve safety, efficiency, resilience, and public service.

Young also connected technological change to education and workforce development. NEIT's role is to prepare future computer scientists, engineering technologists, construction professionals, and other innovators who will help develop and manage increasingly technology-driven infrastructure systems.

Robert Rocchio, P.E. — Interim Director, Rhode Island Department of Transportation

Director Rocchio welcomed attendees on behalf of RIDOT and framed AI within the larger challenge of becoming a modern transportation agency. He emphasized that modernization involves more than investing in infrastructure; it also requires using data effectively to make better decisions, maintain assets, improve safety, and operate the transportation system more efficiently.

Transportation agencies now have unprecedented amounts of information about bridge and pavement conditions, traffic, and changing traveler needs. AI offers tools for turning those data into useful and timely decisions.

Rocchio described a practical pathway for AI adoption: research → understanding what works → pilots → everyday operations. Rather than assuming every AI application will be valuable, agencies should determine where technology produces tangible benefits and then move promising applications into operational use. He noted that the symposium will include examples of AI being used to analyze asset condition information, identify potential problems earlier, and improve decision-making. He suggested that RIDOT's experience had demonstrated surprisingly strong AI capabilities for analyzing and predicting infrastructure issues.

He also cautioned that agencies need to understand AI technology and use it responsibly while identifying applications that can make transportation safer, more reliable, and more efficient.

A particularly important opportunity is addressing transportation workforce constraints. Rocchio noted the difficulty agencies face in recruiting and retaining experienced technical employees. AI can potentially perform repetitive activities and free limited professional staff to spend more time on work requiring expertise, judgment, problem-solving, and creativity.

DAY 1 · SESSION 02

The Road Ahead: AI as the New Infrastructure – FHWA Administrator McMaster

Day 1 Keynote and Fireside Chat

MODERATOR **Pamela Cotter, Administrator of Planning, RIDOT**

SESSION DESCRIPTION

FHWA Administrator Sean McMaster delivered the symposium's opening keynote, followed by a fireside discussion and audience Q&A focused on the role of artificial intelligence in transportation.

SESSION TAKEAWAYS

- **AI offers the potential to improve the performance of the transportation network.** Agencies cannot build their way out of every transportation problem. Technology and AI can increase the value, performance, and efficiency of existing and future infrastructure.
- **AI can help transportation move from reactive to predictive management.** Safety, incident management, and asset management were prominent examples. AI can use multiple data streams to identify elevated crash risk, detect infrastructure deterioration, and anticipate problems before they become emergencies.
- **Rural applications deserve particular attention.** Asset management, planning, emergency response, safety analysis, and eventually automated mobility could provide substantial benefits in rural areas, where distances to services are greater and roadway fatalities are disproportionately high.
- **Some of AI's most immediate benefits may be in everyday agency work.** McMaster highlighted the ability of large language models (LLMs) to analyze lengthy documents, organize public comments, support regulatory processes, review grant applications, and accelerate other information-intensive workflows.
- **Human oversight remains essential.** AI may perform a substantial portion of initial analysis or drafting, but people must review and verify outputs before decisions are made.
- **FHWA wants to help move the industry from individual pilots to scalable deployment.** The planned Center of Excellence, pooled fund study, national roadmap, and portal are intended to share successful applications, establish standards, support research and governance, and prevent agencies from repeatedly developing the same solutions independently.
- **AI literacy is likely to become a basic transportation workforce competency.** McMaster compared future AI proficiency to familiarity with Microsoft Office: employees across disciplines will be expected to know how to interact with AI tools, develop effective prompts, critically evaluate outputs, and use the technology responsibly.
- **AI applications should be accessible to agencies at different levels of readiness.** FHWA intends to support early adopters, mainstream users, and resource-constrained state and local agencies.
- **Recognize that AI in transportation is still at an early stage of implementation.** McMaster characterized federal AI implementation as being in its early stages. The preferred model is to pilot applications at small scale, determine what works, correct problems, and only then expand successful approaches agency-wide.

Sean McMaster – FHWA Administrator

Keynote Presentation

TRANSPORTATION PRIORITIES AND THE ROLE OF TECHNOLOGY

McMaster began by emphasizing the fundamental importance of roads and bridges to everyday life and the economy. He described FHWA's current approach as “back to basics,” focused on building, maintaining, and enhancing infrastructure while helping states and local governments deliver the projects they identify as priorities.

He described project delivery as a central FHWA priority and cited agency statistics regarding grants, obligations, construction projects, and reductions in environmental review times. He then pivoted to the central theme of the symposium: physical infrastructure alone cannot solve every transportation problem. Technology deployment and investment—including ITS, roadway digital infrastructure, and AI—will be essential to improving the performance of the transportation network.



McMaster sees AI as helping create a more predictive and proactive transportation system. Combining ITS applications, digital infrastructure, and AI could improve safety, congestion management, commuting, and system operations while enabling applications not currently possible.

Sean McMaster, Administrator, Federal Highway Administration — Day 1 keynote speaker.

FHWA AI INITIATIVES

A major portion of McMaster's remarks concerned the institutional infrastructure FHWA is developing to support AI deployment. He described four closely related initiatives:

- AI Deployment Center of Excellence **(AI COE).** The AI COE is intended to serve as a national resource to guide AI adoption across surface transportation It will provide interoperable tools, resources, and a trusted repository of best practices and address research, technology, governance, workforce readiness, proven use cases, and deployment support.
- National AI Transportation Coalition Pooled Fund Study**.** A collaborative effort involving transportation agencies and other partners intended to accelerate the adoption of trustworthy and impactful Artificial Intelligence (AI) technologies across the nation's transportation system.
- **Five-Year National AI Transportation Roadmap.** Intended to align federal and state goals and transition transportation from fragmented local pilots toward coordinated, enterprise-scale deployment.
- **National AI Portal.** Envisioned as a single-entry point where state DOTs, local agencies, and industry can access vetted models, standards, research, operational tools, and successful applications.

McMaster stressed that a national roadmap has little value unless it results in resources that practitioners can actually use. A central objective is therefore to make a successful solution developed in one jurisdiction readily available to others rather than requiring each agency to start from scratch.

He later elaborated that the Center of Excellence should make proven use cases effectively “plug and play” and provide a central location for sharing innovations that otherwise might remain isolated within individual agencies.

AI FOR SAFETY AND TRAFFIC OPERATIONS

McMaster highlighted the potential for AI to shift transportation safety from retrospective analysis toward prediction and prevention. He described an FHWA effort using Waze data, weather, historical crashes, and roadway information to generate hourly risk predictions through an interactive dashboard. The resulting safety risk model substantially outperformed benchmark methods and could be used to warn travelers of hazardous situations.

During the Q&A, he cited Utah DOT's AI-driven crash prediction and safety analytics system, which combines crash history, roadway geometry, weather, and traffic flow to calculate relative risk. This helps engineers to identify locations for treatments such as rumble strips, cable barriers, and signal improvements.

He also cited Missouri DOT's use of machine learning to allow Transportation Management Center (TMC) personnel to detect incidents more quickly. Earlier detection can reduce incident duration, secondary crash exposure, and corridor delay and can support proactive measures such as targeted patrol staging and traveler warnings.

AI AND ASSET MANAGEMENT

McMaster identified asset management as one of AI's most promising transportation applications, particularly as the Interstate Highway System and other infrastructure continue to age.

Historically, asset management has depended heavily on manual surveys, visual inspections, and reactive maintenance. AI creates opportunities to automatically detect pavement distress, track bridge deterioration, inventory traffic control devices, analyze inspection imagery, and predict deterioration or failures.

He cited examples involving predictive pavement management and the use of machine learning and computer vision to process large volumes of bridge inspection imagery. The larger objective is to enable earlier, more cost-effective interventions and stretch limited transportation dollars by making better decisions about when and where to maintain infrastructure.

AUTOMATED AND CONNECTED TRANSPORTATION

McMaster also discussed the use of AI in automated vehicle (AV) deployment. He argued that connected and automated vehicles could improve mobility for people unable to drive and have the potential to reduce roadway fatalities and transportation worker risk exposure—for example, through automated construction vehicles.

He emphasized that AV deployment requires thinking about the entire ecosystem: roadway infrastructure, traffic management systems, digital platforms, and machine-readable traffic regulations all need to function together. FHWA is therefore examining how roads are sensed and interpreted by automated vehicles, how traffic control devices should accommodate them, and how traffic laws can be organized into digital formats.

AI FOR GOVERNMENT AND PROJECT DELIVERY

McMaster highlighted AI's potential to transform administrative work. Government agencies generate and review enormous volumes of documents including regulations, environmental materials, grant applications, and public comments. LLMs can perform initial drafting, sorting, categorization, and analysis in seconds or minutes rather than weeks or months.

He gave examples such as developing an initial rulemaking draft, categorizing hundreds of public comments, analyzing lengthy environmental documents, and eventually reviewing and structuring

large numbers of grant applications. His expectation is not that AI will replace professional review, but that it will perform much of the initial processing so employees can concentrate on tasks requiring judgment and substantive analysis.

WORKFORCE DEVELOPMENT

McMaster described AI workforce development as both an immediate training challenge and a longer-term shift in what transportation agencies will expect from employees.

FHWA is developing internal training opportunities, recruiting AI expertise for the Turner-Fairbank Highway Research Center and AI COE, and considering how AI skills will factor into future hiring. He predicted that basic AI literacy—including prompting, understanding appropriate applications, and evaluating outputs—will become a necessary professional competency at DOTs..

He cited five competencies from a Department of Labor AI literacy framework: understanding AI principles, identifying practical uses, directing AI through quality inputs, critically evaluating outputs, and using AI responsibly. FHWA's AI COE is expected to translate these broader AI competencies into transportation-specific tools and training appropriate for different audiences ranging from field engineers to executives.

Fireside Chat and Q&A

Pamela Cotter, RIDOT Administrator of Planning moderated a fireside chat with Administrator McMaster. McMaster posed the first question to Cotter:

WHAT SHOULD BE THE IMMEDIATE IMPACT OF THE SYMPOSIUM, AND WHERE COULD RIDOT BE IN A YEAR?

Cotter explained that the agency already possesses substantial Waze, pavement, bridge, and other data but faces the challenge of making sense of the volume of information. RIDOT has begun pilots and anticipates that the symposium will help the agency understand what other DOTs are doing and identify additional opportunities. As a relatively small DOT, RIDOT is especially interested in using AI to reduce repetitive work. Digital twins were identified as one potential area for further exploration and implementation over the next year.

McMaster agreed that government has large amounts of underutilized information. He highlighted secure access to LLMs within USDOT as an opportunity to analyze information at a scale and speed that previously would have been impractical, while reiterating the need for human verification.

WHERE DOES AI SHOW THE MOST IMMEDIATE PROMISE?

McMaster identified two broad areas. Internally, generative AI can help agencies manage document-heavy regulatory, environmental, administrative, and public-comment processes. Operationally, predictive analytics can improve safety, incident response, emergency management, asset management, and infrastructure decision-making.

With respect to safety and operations, AI is being used to shift from waiting for events to occur and then reacting toward identifying risk and acting proactively. For example, Missouri DOT is using machine learning to accelerate incident detection supporting more proactive strategies for TMCs. Utah DOT is advancing crash prediction models to calculate relative risk scores based on crash data, weather, and road characteristics.

With respect to asset management, McMaster noted that we just celebrated the 70th anniversary of the Interstate System. AI can be used to stretch available dollars for maintaining the system, through automated detection of pavement distress and prediction of structural failures. He noted that Texas DOT's AI strategic plan identified over 230 use cases for AI in transportation including pavement management. Michigan DOT has been using AI to manage bridge inspection data, including post-processing of images for feature identification.

HOW CAN STATE DOTS MOVE FROM PILOTS TO AGENCY-WIDE DEPLOYMENT?

McMaster said state DOTs and AASHTO have already been leaders in transportation AI applications. FHWA plays a complementary role—helping identify and vet successful applications, establish standards, share knowledge, and make proven approaches easier for other agencies to adopt.

The AI COE, pooled fund study, five-year roadmap, and portal are intended to form the institutional framework for AI adoption. McMaster emphasized that smaller local agencies need access to the same knowledge and tools as larger organizations.

HOW IS FHWA PREPARING THE WORKFORCE FOR AI ADOPTION?

McMaster said FHWA is simultaneously training existing staff, recruiting specialists, and preparing for AI competency to become a standard expectation for future employees. The AI COE will also support role-based training for transportation practitioners nationally.

A particularly important point was that workforce preparation involves more than learning how to operate AI tools. Employees must also learn to critically evaluate AI-generated information and use it responsibly.

HOW WILL FHWA BUILD CONFIDENCE AND TRUST IN AI?

McMaster pointed to FHWA's existing Every Day Counts (EDC) model for accelerating adoption of proven innovations. Rather than asking every agency to embrace an untested technology simultaneously, FHWA can identify applications that have demonstrated results, share those experiences through peer exchanges and technical assistance, and encourage broader adoption.

He suggested AI could become a major component of a future EDC cycle. He also described an AI and Transportation Capability Maturity Model (CMM) being expanded by the ITS Joint Program Office to cover areas such as asset management, planning, project delivery, and administration. This could provide an evidence-based way for agencies to assess readiness and determine appropriate next steps.

WHAT OPPORTUNITIES DOES AI OFFER RURAL TRANSPORTATION?

An audience member asked whether the AI discussion was primarily focused on urban congestion and highway operations. McMaster responded that many applications—particularly asset management, planning, infrastructure design, and administration—are equally applicable in rural settings.

He highlighted emergency response and safety as areas where the benefits could be particularly significant because emergency services are often farther away and rural roads account for a disproportionate share of roadway fatalities. He also sees automated mobility eventually providing important access for older adults, people with disabilities, and others in rural areas who cannot drive, although he expects AV deployment in rural communities to lag urban deployment.

WHAT ABOUT ADMINISTRATIVE FUNCTIONS SUCH AS RECORDS MANAGEMENT?

The final audience question broadened the discussion beyond engineering and operations to records management and other administrative functions necessary to run a DOT.

McMaster acknowledged that adoption of AI for these applications within federal agencies is currently limited but he sees substantial potential. Digitizing records and processes creates the foundation for AI to analyze documents, connect information across systems, streamline grants management, organize applications, and answer questions that currently require significant manual effort.

He described FHWA as deliberately testing AI in small-scale applications rather than immediately deploying systems agency-wide. The approach is to identify a manageable use case, test the technology, determine whether outputs are reliable, address problems, and then scale successful applications. He predicted that within two to three years the conversation about AI-enabled federal workflows could look substantially different.

DAY 1 · SESSION 03

Governance Frameworks for Successful AI Development and Deployment in Transportation

Panel Discussion 1

MODERATOR **Lawrence Dwyer, Director of Safety and Mobility, AASHTO**

SESSION DESCRIPTION

This panel focused on what transportation agencies need to put in place to move from experimenting with AI to deploying it responsibly and sustainably. The session moderator framed governance, transparency, accountability, procurement, data stewardship, privacy, fairness, liability, and public trust as issues that transportation leaders need to address now—not at some future point after AI has been deployed. Panelists represented state government, state DOT, federal, research, and industry perspectives.

SESSION TAKEAWAYS

- **Governance needs to be tailored to different types of AI:** Maturity levels and risk profiles vary across traditional machine learning and numerical pattern recognition; computer vision and image recognition; and generative AI.
- **Responsible AI deployment is less about buying the right technology than building an organization capable of making good decisions about technology.** Therefore, AI governance needs to become part of transportation agencies' broader approach to leadership, workforce development, data management, procurement, risk management, budgeting, and organizational change.
- **AI governance is primarily an organizational and people challenge, not a technology challenge.** Successful deployment requires a collaborative effort involving executive leadership, program staff, IT, data, legal, procurement, communications, change management, and end users.

- **Start by defining the problem to be solved with AI.** Agencies should clearly define why they are considering AI and what transportation or business problem they are trying to solve before selecting a technology.
- **Governance does not need to be perfect before agencies begin experimenting.** Agencies can pilot AI while developing governance, but they should establish decision structures, responsibilities, risk controls, and processes early rather than trying to retrofit them after deployment.
- **Trust is fundamental.** Employees and the public need confidence in the data, processes, safeguards, and people responsible for AI. Transparency about why AI is being used and how decisions are made is an important part of establishing and maintaining that trust.
- **Good data is foundational.** AI is only as useful and trustworthy as the data behind it. Data governance and AI governance therefore need to be closely connected.
- **Workforce development is part of governance**. Training should extend beyond AI specialists to managers, program staff, lawyers, procurement staff, communications professionals, and other employees who will interact with AI or make decisions about it.
- **Keep humans "at the helm."** The panel went beyond the common phrase "human in the loop": emphasizing that people remain responsible for reviewing, accepting, and acting on AI outputs.
- **Governance should be risk-based**. A safety-critical AI application requires much stronger testing, verification, validation, and oversight than a low-risk administrative application.
- **Pilots should demonstrate value before agencies scale.** MnDOT's approach emphasizes cautious experimentation and avoiding large investments in untested technologies.
- **AI requires changes to processes and budgets, not just new tools**. Agencies should reconsider existing workflows to capture AI's benefits and begin treating digital capabilities as core infrastructure requiring sustained funding.

Scott Gausland — Chief Data and Analytics Officer, State of Rhode Island

Using AI to Move from Asset Data to Projects

Gausland, Rhode Island's Chief Data and Analytics Officer, described the state's emerging governance structure. Rhode Island is creating a Data Analytics and AI Center of Excellence that will serve as an innovation hub and advisory resource for agencies considering AI projects.

Currently, Rhode Island uses its existing data and IT governance process to approve AI projects. The Center of Excellence is intended to expand that approach by evaluating proposed AI projects through a formal framework, helping agencies develop use cases, and bringing together talented staff from across state government.

An important feature of the model is that the state does not intend the Center of Excellence to be composed solely of outside AI specialists. Gausland wants to develop and upskill employees already working in state agencies because they understand the programs, problems, and operational context. The goal is to build an internal community capable of developing AI and analytics solutions around real agency needs.

He also emphasized ongoing model validation. Governance cannot end when an AI system is approved and deployed; models need to be periodically checked to determine whether they continue to perform as intended and remain aligned with their original purpose.

Rhode Island is simultaneously developing a statewide data governance program and integrated data system. Gausland sees a common platform, common tools, and shared skill development across agencies as the foundation for more advanced analytics and AI.

In the discussion, his central governance lesson was collaboration. He cited RIDOT's early engagement with statewide IT/data governance staff on an AI project: because the governance team was involved early, many questions had already been addressed by the time the project reached formal review.

His final advice was to focus on organizational change and process redesign. Agencies should not simply layer AI onto existing processes; they should ask how the process itself could work better and then determine how AI can support that change.

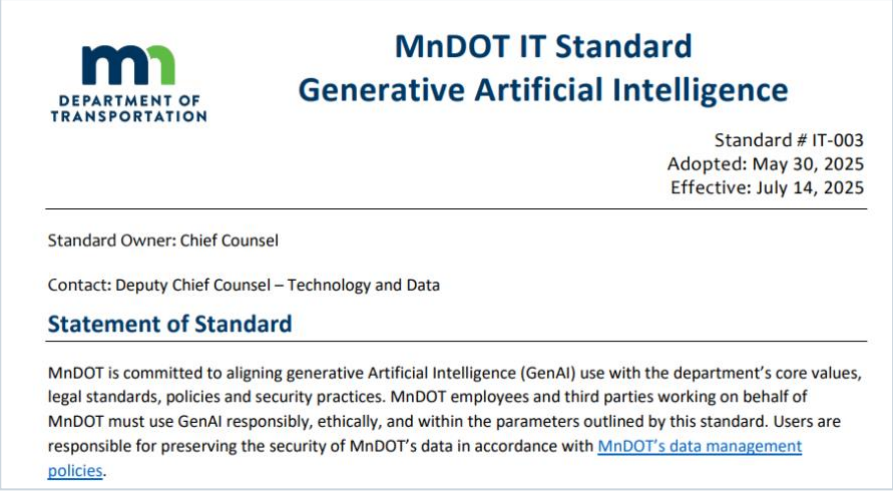
Jocelyn Stein — Business Integration & Digital Accessibility Lead, Minnesota Department of Transportation

MnDOT AI Experiences

Stein described MnDOT's AI journey as an effort at the intersection of technology, people, and processes.

MnDOT has had an AI work group for several years. It is led by the Office of Chief Counsel and reports to the Technology Data Leadership Committee, which supports enterprise IT and data strategy. The agency also has AI pilots underway in asset management, traffic management centers, freight and commercial vehicle operations, crash analytics, and other areas.

MnDOT has established a Generative AI Standard, currently supports Microsoft Copilot, and was evaluating Claude at the time of the symposium. The agency is also participating in the statewide "AI the Minnesota Way" initiative, intended to develop a broader cross-sector vision for AI in Minnesota.



DEPARTMENT OF TRANSPORTATION

MnDOT IT Standard
Generative Artificial Intelligence

Standard # IT-003
Adopted: May 30, 2025
Effective: July 14, 2025

Standard Owner: Chief Counsel

Contact: Deputy Chief Counsel – Technology and Data

Statement of Standard

MnDOT is committed to aligning generative Artificial Intelligence (GenAI) use with the department's core values, legal standards, policies and security practices. MnDOT employees and third parties working on behalf of MnDOT must use GenAI responsibly, ethically, and within the parameters outlined by this standard. Users are responsible for preserving the security of MnDOT's data in accordance with MnDOT's data management policies.

Minnesota DOT's AI governance structure and standards, presented by Jocelyn Stein.

A particularly strong part of MnDOT's approach is workforce training. The agency developed multiple levels of Copilot training; Stein reported that more than 18% of MnDOT's approximately 5,300 employees had already taken at least one training course. Training covers not only tool use but guardrails and expectations involving privacy, bias, fairness, and transparency. MnDOT's message to employees is that AI is an assistant and that a human remains in the loop.

MnDOT also surveyed employees and found a wide range of attitudes—from excitement about improved productivity and customer service to concerns about bias, sustainability, and other risks.

The agency's next steps include standardized AI-related contract language developed by the Chief Counsel, an agency-wide AI vision and roadmap, continued pilots, and an AI champions program to support enthusiastic early adopters without forcing adoption on employees who are not ready.

For Stein, the most important governance challenge is trust—trust among employees, the public, other stakeholders, and trust in the underlying data. She characterized AI adoption as fundamentally a people and change-management issue rather than simply a technical one.

Her advice to transportation executives was correspondingly cautious: don't try to boil the ocean. Pilot promising applications, learn from them, maintain momentum, but avoid committing substantial resources to systems that have not yet been tested and proven.

Lisa Randall — AI Program Manager, USDOT ITS Joint Program Office

Bridging AI Gaps for Transportation Agencies: ITSJPO AI Program & Products

Randall offered a broader framework for thinking about AI governance and stressed that “AI is a team sport.” Governance structures need to engage executives, middle managers, IT and data staff, legal counsel, program specialists, procurement professionals, and others rather than leaving AI to technologists.

She identified several major components of effective governance:

- **Organizational structure and decision rights.** Agencies need some form of executive-level decision-making body, clearly defined roles, oversight responsibilities, and decision gates.
- **Risk management.** Governance should address both enterprise-level AI risk and risks associated with individual projects. The rigor of oversight should correspond to the use case; safety-related applications require particularly strong testing, evaluation, verification, and validation.
- **Procurement.** Purchasing AI and other rapidly changing technologies differs substantially from traditional capital procurement. Agencies need to consider AI throughout the pre-award, award, and post-award lifecycle.
- **Needs analysis**. Before pursuing an AI project, agencies should ask why AI is needed. Is an existing process too expensive or slow? Are important data sources not being integrated? Is another agency's successful application genuinely transferable?

Randall cited Florida DOT as an example of a more centralized approach. FDOT has developed a process for collecting AI ideas, applying criteria, having a cross-unit group prioritize them, and selecting projects for funding and testing.

She also emphasized workforce and change management. Agencies need not only data scientists but communicators who can explain AI internally and externally and change-management specialists who can help employees understand the benefits, risks, and organizational implications.

USDOT/ITS JPO has established an internal AI community of practice and AI literacy activities. JPO is also developing resources including AI governance guidance, peer examples, an AI case-study series, and updates to its AI capability maturity model. The planned case-study resource

included approximately 30 standardized case studies covering tribes, transit agencies, local agencies, and state DOTs.

Randall's emphasized the importance of “defining your why for AI.” Determine why AI is being considered and why it could help solve a specific transportation problem; the what, how, and who should follow from that.

Her closing advice focused on the workforce and on moving beyond “human in the loop” toward “human at the helm”—humans remain the decision-makers responsible for reviewing and accepting AI-generated outputs.

Matthew Cuddy — Innovation Manager and AI Applications Strategist, Volpe National Transportation Systems Center

Cuddy reinforced the workforce theme but focused particularly on the pace of technological change. Agencies cannot expect one centralized office to understand, screen, and evaluate every new AI tool. Instead, they need an organizational training and learning system that equips employees throughout the agency to experiment appropriately, understand emerging tools, and make informed decisions.

His vision was essentially an agency-wide innovation system in which employees in different functions have enough AI understanding to fulfill their respective roles. This is particularly important because the AI market is evolving so rapidly.

Cuddy's final advice introduced an important distinction: executives should be clear about what kind of AI they are discussing. He distinguished the following three categories:

- traditional machine learning and numerical pattern recognition;
- computer vision and image recognition; and
- generative AI.

These technologies have different maturity levels and risk profiles. Governance principles may apply across all three, but the specific controls, expectations, and deployment strategies should differ. Generative AI in particular remains less mature and can hallucinate, so it should not automatically be governed or deployed in the same way as well-established machine-learning or computer-vision applications.

Laura Chace — President and CEO, ITS America

Chace presented an industry perspective based on ITS America's work with public agencies, private companies, and academia.

Her most important framing was that AI implementation is not fundamentally a technology challenge—it is an organizational challenge. Agencies should resist starting with “we need to deploy AI” and instead begin with the problem they are trying to solve. AI may ultimately be the right tool, but technology selection should follow problem definition.

ITS America developed a practical AI implementation guide with Cambridge Consultants in response to requests from public-sector members. The framework divides AI governance into three interconnected functions:

- Executive function — strategic direction and leadership.

- Operational function — foundations and translation of strategy into how work gets done.
- Delivery function — implementation.

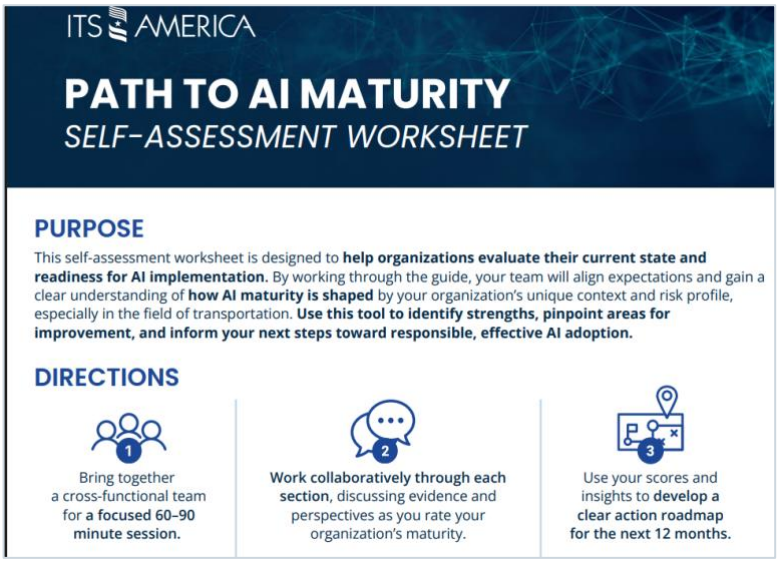
All three need to work together; otherwise agencies are likely to encounter friction as they try to scale AI.

ITS America also developed a path-to-AI-maturity self-assessment that allows agencies to determine where they are on a maturity continuum and identify practical next steps rather than expecting every organization to immediately develop a comprehensive AI program.

Chace placed responsible AI at the center of governance. That means communicating what agencies are doing internally and publicly, understanding how systems reach decisions, maintaining transparency, determining responsibility when systems produce incorrect results, and explicitly addressing privacy, security, and bias.

She also strongly emphasized data: AI systems cannot compensate for poor underlying information. Agencies need to invest in getting their data into a condition suitable for AI use.

For Chace, the most important governance challenge was transparency, encompassing privacy and security as well as communication with employees and the public. When organizations fail to explain why and how AI is being used, people may assume the worst.



Responsible AI at the center of governance, presented by Laura Chace, ITS America.

Her closing recommendation went beyond governance: rethink budgets. Digital tools have become part of agencies' core infrastructure, but transportation budgeting practices have not necessarily evolved accordingly. Executives need to plan for sustained investment in the digital systems that increasingly operate transportation organizations and networks.

Discussion and Q&A

Dwyer asked each panelist to identify the single most important governance challenge agencies need to solve over the next three to five years.

- **Jocelyn Stein / MnDOT**: Trust — with employees, the public, stakeholders, and in the underlying data
- **Matthew Cuddy / Volpe**: Workforce learning and training — equip the organization to keep pace with a rapidly changing technology landscape
- **Lisa Randall / USDOT JPO**: Define the "why" — start with the problem and intended outcome

- **Scott Gausland / Rhode Island**: Collaboration — bring governance, technology, program, and business staff together early
- **Laura Chace / ITS America**: Transparency — explain why and how AI is being used and address privacy and security

Dwyer highlighted the importance of having a strategic framework to guide AI efforts and establish an overall plan before isolated use cases proliferate. He specifically cited Texas and Vermont as examples of agencies with useful strategic frameworks. He observed that organizations that begin with one isolated AI application can quickly encounter governance, legal, data, workforce, and policy questions they had not anticipated. Having a strategic framework in place makes those questions easier to address.

Dwyer posed a final question to the panel: what is the one piece of advice you would give to transportation executives on AI governance and deployment. Responses were as follows:

- **Lisa Randall / USDOT JPO**: Focus on building workforce capacity – they are a critical part of the process (not just tech specialist – IT, legal, procurement, data have a piece of the AI puzzle.)
- **Matthew Cuddy / Volpe**: Be thoughtful and clear about what type of AI you are planning for – this is a class of technologies – ML, computer vision, GenAI. They are at different points in their maturity.
- **Jocelyn Stein / MnDOT**: Don't need to boil the ocean – many things you can do. Don't get ahead of yourselves, do pilots, be cautious, prove before jumping ahead.
- **Scott Gausland / Rhode Island**: Include Organizational Change Management (OCM) to get the most value from AI – rethink processes, how can we improve them using AI? Make sure folks are on board with that.
- **Laura Chace / ITS America**: Recognize that digital tools are core infrastructure – rethink how we are planning and budgeting for those. Think about what a family spends on technology now vs in the 1980s.

DAY 1 · SESSION 04

From Reactive to Predictive: Using AI to Advance Transportation Safety

Technical Session 1

MODERATOR **Lawrence Dwyer, Director of Safety and Mobility, AASHTO**

SESSION DESCRIPTION

This session examined how artificial intelligence, machine learning, computer vision, and new data sources can help transportation agencies move from a largely reactive safety model—waiting for crashes to occur and then analyzing them—to a more predictive and proactive approach. Dwyer noted that traditional crash data can lag by a year or more and may have accuracy limitations, while emerging technologies can identify risky conditions, behaviors, and near misses much closer to real time.

The session combined two applied examples with a broader research perspective. Gregory Ciparelli of Connecticut DOT described a University of Connecticut computer-vision pilot that uses video to analyze intersection operations and near misses. Daniel (Mac) Lang of the USDOT Volpe

Center described an AI rapid-prototyping effort with MnDOT and Washington State DOT to predict roadway risk using fused data. Katherine Kortum of the Transportation Research Board placed those applications in the broader context of predictive safety research and emphasized the importance of validation, understanding model failures, and—most importantly—turning predictions into actual safety improvements.

SESSION TAKEAWAYS

- **Start with the safety problem, not the AI tool.** Agencies should first determine what they are trying to impact—crashes, near misses, unsafe behavior, operational risk—and only then select an analytic approach that supports what they are trying to do.
- **AI is expanding the data sources available for safety analysis.** Crash records are no longer the only—or necessarily the most useful—source for identifying risk. Video, computer vision, probe data, crowdsourced information, weather data, LiDAR, and connected-vehicle data can reveal conflicts and risky behaviors before crashes occur.
- **Near-miss data may be one of the most valuable new opportunities.** Computer vision can identify interactions that previously disappeared without leaving a record—fast turns, pedestrian hesitation, evasive bicycle movements, vehicle-pedestrian proximity, and other conflicts.
- **AI can support both operational and strategic safety decisions.** Traffic-management operators may need minute-by-minute risk information, while planners and safety staff may need broader patterns across days, months, or years. The same underlying analytics may therefore need different interfaces and outputs for different users.
- **Risk prediction is more practical than attempting to predict individual crashes.** Because crashes are relatively rare events, the Volpe pilot focused on producing relative risk scores for roadway segments rather than predicting that a specific crash would occur at a specific place and time.
- **Real-world pilots are essential.** Rapid prototyping allows transportation agencies to test whether promising AI approaches actually integrate into existing operations, workflows, data systems, and staff practices before committing to a major procurement.
- **AI augments rather than replaces human expertise**. Operators need to understand the system, know what an alert means, and have predefined action thresholds and procedures.
- **Models must be locally validated and continually monitored.** A model that performs well in one geography or season may not perform equally well elsewhere or later. Agencies need to watch for model drift and changing conditions.
- **High overall model accuracy is not enough for safety-critical applications.** Transportation agencies need to understand where and why a model fails, and what consequences those failures could have.
- **Better prediction is only the beginning.** Even a perfect model does not make a roadway safer unless the agency uses its findings to select, fund, implement, and evaluate countermeasures. Agencies must validate models, understand failure conditions, tailor information to different users, retain professional judgment, establish clear action protocols, and ultimately connect AI findings to decisions and investments that reduce fatalities and serious injuries.

Gregory Ciparelli — Transportation Planning Director and Chief Data Officer, Connecticut Department of Transportation

AI-Based Video Analytics

Ciparelli presented research conducted by the University of Connecticut in partnership with Connecticut DOT on AI-based video analytics for intersection safety and operations. Connecticut DOT has used university research partnerships as an important avenue for testing AI applications before deploying them directly on DOT infrastructure.

COMPUTER-VISION PROOF OF CONCEPT

The initial project tested whether relatively inexpensive cameras and computer vision could extract detailed operational and safety information from signalized intersections on the UConn campus.

The university installed custom camera equipment costing less than $100 per unit at three intersections. The system used computer vision to identify and track vehicles and pedestrians and derive information including vehicle classifications and trajectories, speeds, traffic volumes by mode, turning movements, pedestrian movements; and interactions among road users.

Rather than simply producing turning-movement counts, the technology followed individual road users through the intersection and captured information that a traditional manual count would not provide.

Ciparelli emphasized the contrast with traditional field data collection. Human observers can count movements, but fatigue and boredom affect data quality, and manual counts generally do not capture speed, trajectory, or detailed conflict information. The pilot demonstrated that computer vision could collect substantially richer information.

NEAR-MISS AND CONFLICT ANALYSIS

The project was particularly interested in measuring **near misses** involving vehicles, pedestrians, and other vulnerable road users. Two surrogate safety measures were highlighted:

- **Time to Collision (TTC)** estimates the time remaining before a collision would occur if road users continued on their current trajectories. Shorter TTC values indicate less time available for braking or evasive action.
- **Post-Encroachment Time (PET)** measures the time between one road user leaving a conflict point and another arriving at that same point. Values approaching zero indicate increasingly serious conflicts.

The analysis identified numerous short TTC and PET events, including vehicle-pedestrian interactions. Ciparelli emphasized that these surrogate safety measures provide safety engineers with information they previously did not have and potentially allows agencies to intervene before a location accumulates a significant crash history.

MOVING TO REAL-TIME ANALYSIS

The successful proof of concept helped lead to a $1.57 million Safe Streets and Roads for All grant to expand the approach.

The next phase will use four intersections, 5G connectivity, and a real-time dashboard. Vehicle re-identification will allow the system to recognize an anonymous vehicle signature as it moves from camera to camera without capturing a license plate or registration information.

This could allow analysts to distinguish between repeated unsafe behavior by one particular vehicle and a recurring infrastructure or operational problem affecting many vehicles at the same location. That distinction can help determine whether a problem is principally behavioral, engineering-related, or some combination of the two. Future outputs are expected to include near-miss heat maps, safety performance measures, and additional mobility and safety analytics.

Daniel (Mac) Lang — Policy Analyst, USDOT Volpe National Transportation Systems Center

AI Rapid Prototypes: Predictive Analytics for Incident Management

Lang discussed the **ITS Joint Program Office's AI Rapid Prototyping Initiative**, which seeks to shorten the path between transportation AI research and practical deployment.

He noted that conventional research cycles can be poorly matched to AI because technologies can change significantly by the time a traditional study is completed and published. Rapid prototyping provides an opportunity to test technologies directly with state and local agencies, obtain operational feedback, and determine what actually works in practice.

The first round included four pilots, with additional pilots planned. His presentation focused on an incident prediction and management prototype tested with MnDOT and Washington State DOT.

FUSING MULTIPLE DATA SOURCES

The project combined several sources to produce a roadway risk metric: Waze Data for Cities information on crashes and congestion; historical crash data; historical and forecast weather information; and roadway geometry and roadway type.

Waze information proved particularly valuable because it provided more current information than traditional crash databases.

The model generated an hourly relative risk score for roadway segments over approximately the next three to five days. Results were displayed on an interactive dashboard, allowing users to examine locations and predicted risk over time. Identification of segments with higher relative risk provides valuable information to help agencies prioritize attention and resources for maximum impact.

MINNESOTA DOT PILOT

The MnDOT pilot involved traffic-management-center operators and produced encouraging statistical results. According to Lang, the model identified high-risk corridors 12 to 22 times better than a naïve baseline.

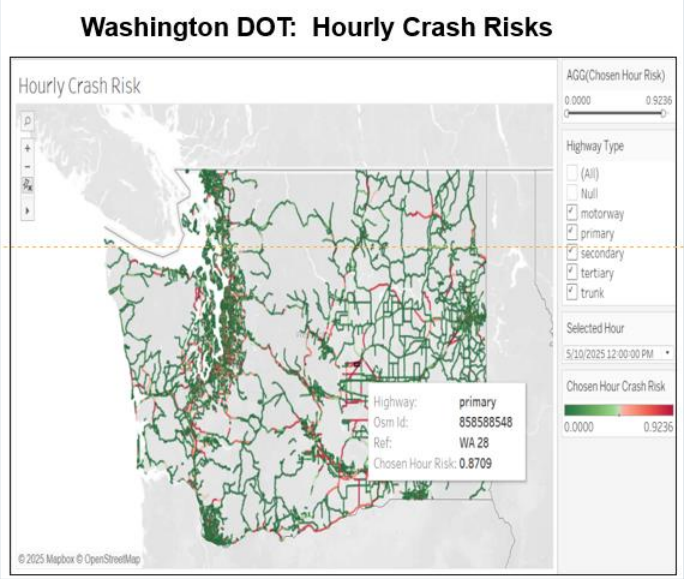
Model performance was strongest during weekday daytime and peak periods, and on higher-volume roadways such as Interstate, U.S., and major state routes. Model performance declined during atypical periods such as holiday weekends, demonstrating the need to account for holidays, special events, and other conditions that differ from normal traffic patterns.

WASHINGTON STATE DOT PILOT

The Washington State DOT application focused more on planning than real time operations.

Staff found that the risk hotspots produced by the model generally aligned with their own understanding of the system.

That suggested another potentially useful application: capturing institutional knowledge. A risk map can help new staff understand known high-risk corridors without relying entirely on experienced employees to transfer that knowledge individually. The tool could also help agencies prioritize countermeasures and deployments when resources are limited.



WSDOT's crash-risk hotspot mapping, presented by Mac Lang .

LESSONS FROM DEPLOYMENT

Lang emphasized several lessons from the pilots:

- **Train and involve the workforce.** Employees need familiarity with and confidence in the system. AI should supplement staff knowledge, not override it.
- **Link predictions to action.** An alert is only useful if an operator knows what threshold has been crossed and what response is appropriate.
- **Design tools around users.** TMC operators want streamlined alerts; planners often want detailed data and explanations. One interface does not fit every use case.
- **Find organizational champions.** Willing partners at MnDOT and Washington State DOT helped engage staff and create a productive testing environment.
- **Prototype before procuring at scale.** Pilot programs allow agencies to understand benefits, limitations, and organizational requirements before making large investments.

He also noted that JPO's rapid prototyping efforts can extend beyond highway safety to other multimodal use cases. They are continuing to pursue other prototypes and are open to additional ideas. The prototyping approach provides low risk opportunities for partners to test AI, and participating agencies receive a custom prototype for future use.

Katherine Kortum — Senior Program Officer, Transportation Research Board

Kortum emphasized that the practice of predictive safety itself is not new; what AI is changing is the amount and type of information transportation professionals can observe and analyze.

MAKING THE UNOBSERVABLE OBSERVABLE

Traditional crash analysis faces a fundamental problem: someone has to crash before there is crash data. Individual intersections may experience only a few crashes over several years, which provides little statistical information for determining why the location is risky.

With computer vision and other new data sources, agencies can now observe events indicative of safety risk that were not previously being recorded:

- motorists turning too quickly;
- vehicles cutting corners;
- pedestrians beginning to cross and then retreating;
- bicyclists making evasive movements; and
- road users passing dangerously close to one another.

Kortum summarized this distinction by noting that AI does not necessarily improve safety by itself; rather, it can make previously unobservable safety conditions observable.

DEFINE WHAT "PREDICTION" MEANS

She cautioned transportation practitioners to be precise when they say an AI system "predicts crashes." Agencies need to understand exactly what the model predicts, how a "crash" is defined, and—most importantly—what decision the agency is prepared to make based on the prediction. This point complements Lang's distinction between predicting individual crashes and estimating relative roadway risk.

THE 99% PROBLEM

Kortum's strongest caution concerned model accuracy in safety-critical applications. .A model described as 99% accurate may sound exceptionally successful from a general machine-learning perspective. But an accurate model isn't the same as a safe system. Transportation safety professionals need to understand the remaining 1%—particularly if failures could contribute to death or serious injury. For safety applications, understanding the long tail of failures may be more important than a single overall accuracy statistic. Model failures might reflect impacts of factors such as weather, night driving conditions, occluded camera views, work zones, and recent construction affecting road conditions.

LOCAL VALIDATION AND MODEL DRIFT

Models trained in one setting should not automatically be assumed to work elsewhere. An approach validated in Manhattan may not perform the same way in Providence, Oklahoma City, Sioux Falls, Phoenix, or a snowy rural environment.

Models also need to be checked after deployment. Their performance may change as weather, traffic patterns, infrastructure, or underlying data change. Agencies therefore need to monitor for performance drift, not simply validate the system once at implementation.

PREDICTION IS NOT PREVENTION

Analytics have value only when they lead to action. Even a perfectly accurate model does not save lives on its own. An agency still must determine what is causing the risk; select an appropriate countermeasure; prioritize the location; obtain funding; assign staff; implement the treatment; and finally, evaluate whether conditions improved.

More information can therefore create a new management challenge. If AI identifies 1,000 high-risk locations rather than 10, agencies still have to decide which ones to address first and how to allocate scarce resources.

Kortum noted that there is a workforce component to consider that goes beyond understanding of data science. Staff must be grounded in basic safety professional practices including understanding the data sources, and how they are validated and checked.

Kortum's closing message was that the objective is not a better AI model; the objective is a safer transportation system**.** Seeing risk is not the same as fixing it, and predicting crashes is not the same as preventing them.

Discussion and Q&A

HOW SHOULD AGENCIES BALANCE MOBILITY AND SAFETY IN AI-DRIVEN OPERATIONS?

Dwyer asked how agencies can balance safety with mobility objectives when deploying technologies such as adaptive signal control, noting that transportation organizations often have separate safety, congestion, operations, and infrastructure silos.

Ciparelli/CTDOT – Ciparelli briefly described Connecticut DOT's adaptive signal-control pilot on Route 15 which will provide an opportunity to explore how to balance safety and mobility objectives. The corridor contains numerous driveways and interconnected signalized intersections, and the agency was still awaiting final pilot results at the time of the symposium.

Lang/Volpe - Lang stated that the accessibility of new data and new ways to operationalize that has encouraged people to think broadly about what kind of questions can be answered and how we can answer them. He emphasized the need for continuing cross-disciplinary communication. New datasets and AI applications often span traditional organizational boundaries, so agencies should actively identify where similar tools are being developed by different groups and whether a common approach could address several problems. Peer exchanges—both within organizations and among agencies—can help uncover these overlaps and expand the range of questions transportation data can answer.

Kortum/TRB – Kortum observed that despite the fact that every DOT will tell you that safety is their "North Star," there continues to be many deaths on the nation's highways every year. Her broader point was that agencies should judge AI applications by whether they actually change safety outcomes, rather than by the sophistication of the technology or the existence of an AI strategy.

HOW CAN A STATE PARTICIPATE IN USDOT'S AI RAPID-PROTOTYPING PILOTS?

An audience member asked how states could participate in future rapid prototyping additional pilots discussed by Lang.

Lang encouraged interested agencies to contact him, the Volpe Center, or ITS JPO Program Manager Lisa Randall and discuss possible alignment between problems the state wants to address, AI applications USDOT wants to research, and opportunities for practical field deployment.

He noted that current work included computer vision, pedestrian detection with the Federal Railroad Administration, and bus-bunching prediction with the Federal Transit Administration, and that the program was actively seeking state and local partners.

DAY 1 · SESSION 05

Optimizing ITS and Mobility Systems with AI

Technical Session 2

MODERATOR **Laura Chace, President and CEO, ITS America**

SESSION DESCRIPTION

This session examined how AI can be used to improve the performance of intelligent transportation systems (ITS) and mobility services. It provided a broad view of how AI may change transportation operations—from proven applications such as signal optimization and vehicle dispatching to future multi-agent systems capable of coordinating an entire transportation network.

Chace opened the session by emphasizing that these applications are already producing measurable benefits. She cited examples of AI reducing the labor required for signal retiming, AI-powered intersections improving pedestrian safety, and automated incident detection reducing emergency response times and secondary crashes. Her framing was that AI's value depends not simply on the technology but on whether it is thoughtfully integrated into transportation operations.

The three panelists approached the topic from complementary perspectives.

- **Lawrence Dwyer of AASHTO** surveyed practical ITS applications already being deployed by state transportation agencies.
- **Jinhua Zhao of MIT** contrasted “grounded AI”—applications embedded in real organizations and human behavior—with more “fantastical” frontier research involving multimodal, agentic, and self-improving AI.
- **Donovan Guin of IBM** focused on the data and information architecture required to integrate otherwise siloed ITS systems and enable more holistic, agentic transportation management.

SESSION TAKEAWAYS

- **AI does not change the fundamental mission of ITS; it changes when agencies can intervene.** Transportation agencies have always sought to move people and goods safely and efficiently, but AI makes it increasingly possible to anticipate congestion, incidents, and operational problems instead of simply responding after they occur.
- **Some of the clearest near-term AI applications are in traffic operations.** Adaptive signal timing, incident detection, predictive congestion management, vulnerable-road-user detection, and traveler information applications are already producing tangible results.
- **Data architecture is integral to successful AI deployment for mobility applications** The panel emphasized that agencies cannot optimize an entire corridor or network when signals, tolling systems, traveler information, traffic management centers, connected vehicles, and other systems remain isolated. As Guin put it, there is effectively "no AI without information architecture (IA)."
- **AI applications are moving from prediction toward coordinated action.** The emerging frontier is not simply identifying that congestion is likely but having multiple systems work together to validate the prediction, evaluate alternatives, adjust signals, inform travelers, reposition resources, and potentially execute a response.
- **Technology alone is not the constraint to advancement.** Human behavior, organizational readiness, data integration, procurement, trust, business-case development, and local operational knowledge determine whether AI produces real value.
- **"Human in the loop" means more than approving an AI recommendation.** Local supervisors and frontline staff possess contextual and organizational knowledge that an algorithm may never see. Their participation can determine whether an AI pilot succeeds at all.
- **High ROI exists at the application level, but enterprise-level ROI remains difficult to prove.** Individual use cases show measurable gains, yet large statewide ITS and AI investments can cost tens or hundreds of millions of dollars. Agencies need to compare benefits such as reduced delay, improved safety, and freight efficiency with competing infrastructure investments.
- **Agentic AI may arrive first in relatively low-risk operational and administrative functions.** Examples discussed included processing 311/511 inquiries, generating traveler-information messages, identifying infrastructure issues from vehicle video feeds, and coordinating information across systems before moving toward more autonomous traffic-management decisions.
- **Transportation agencies should understand and prepare for dramatic advances in agentic and multimodal AI.** However, near-term success will depend on grounding those technologies in the people, processes, data, infrastructure, and institutional realities of transportation agencies.
- **AI capability is advancing faster than governance, procurement, alignment, and institutional adoption.** Government procurement requirements slow deployment but also provide safeguards against adopting rapidly evolving technology irresponsibly.

Larwrence Dwyer — Director of Safety and Mobility, AASHTO

Dwyer set the practical context for the session by arguing that AI has not changed what ITS is trying to accomplish—it has changed how early transportation agencies can act. Traditional operations management strategies have included signal timing, ramp metering, and incident detection. AI increasingly allows agencies to anticipate congestion and non-recurring events rather than waiting until the transportation system has already degraded.

MOVING FROM REACTIVE TO PREDICTIVE OPERATIONS

Dwyer described AI as supporting a transition from response to proactive intervention. This transition can affect congestion management, safety, emergency response, and traveler information. A model might not just forecast where congestion or crashes are likely but recommend where to stage law enforcement or tow trucks, when to activate variable-message signs, and when to provide information to navigation platforms so travelers can adjust routes.

SAFETY AND MOBILITY APPLICATIONS

Computer vision, LiDAR, and other sensors can enable functions such as work-zone intrusion detection; automated speed enforcement; vulnerable-road-user detection; dynamic dispatching of emergency vehicles; demand prediction; route optimization; and automated shared mobility.

STATE EXAMPLES

Dwyer highlighted several examples illustrating the progression toward AI-enabled ITS.

- **Utah DOT** has built one of the country's most extensive signal performance-measurement programs. Approximately 1,800 of the state's roughly 2,000 signals produce continuous performance data. That existing data foundation is now supporting machine-learning-based signal optimization.
- **Maricopa County, Arizona** tested AI-driven adaptive signal control using real-time camera information rather than fixed time-of-day plans. Dwyer reported that a one-week evaluation reduced vehicle intersection delay from 29 to 13 seconds, reduced cross-traffic delay by 54%, and was estimated to reduce operating costs substantially.
- **Georgia DOT's NaviGAtor Advanced Traffic Management System (ATMS)** demonstrates the value of long-term operational data. The system has been operating since the 1996 Olympics and has historically reduced incident duration and millions of vehicle-hours of delay. Georgia is now layering predictive analytics onto the system.

Agencies with mature ITS systems and strong data foundations are particularly well positioned to add AI because they already possess the information needed to train and operate more sophisticated models.

RETURN ON INVESTMENT (ROI) CHALLENGE

During the discussion, Dwyer emphasized how difficult it is for transportation executives to justify large digital investments against investments in visible physical assets.

He cited Iowa's approximately $100 million statewide ITS software replacement as an example. A DOT leader presenting that investment to elected officials must explain why the money should go to ITS rather than perhaps rehabilitating bridges or paving Interstate miles.

The ROI case needs to account for user benefits such as reduced delay, better freight movement, improved safety, and operating efficiency—not simply agency cost savings.

AI can significantly improve transportation operations, but agencies need both a mature data foundation and a credible business case connecting operational improvements to outcomes decision-makers value.

Jinhua Zhao — Professor of Cities and Transportation, MIT

Fantastical AI vs. Grounded AI

Zhao's presentation distinguished frontier research that demonstrates what AI may someday accomplish and AI that has been tested within actual transportation organizations, where human

behavior, institutional constraints, frontline operations, and organizational culture determine whether a technically strong system succeeds.

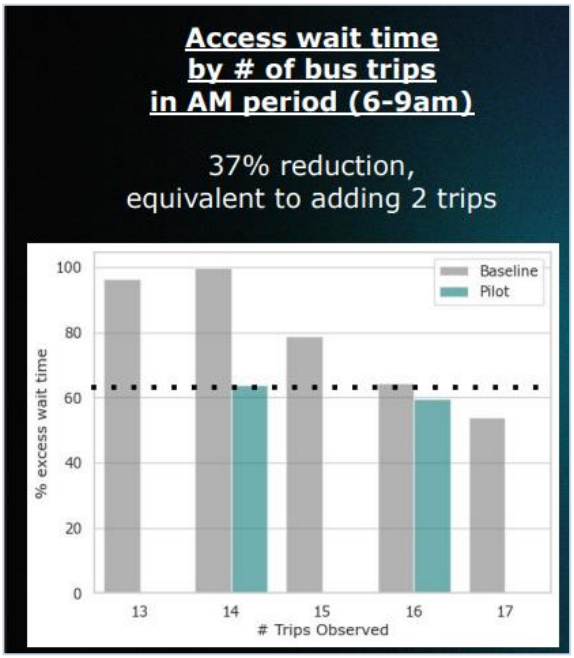
GROUNDED AI

Zhao's principal grounded example involved MIT's work with the **Chicago Transit Authority (CTA)** on bus bunching and service reliability.

The problem became particularly severe during and after COVID because of driver shortages and absenteeism. Zhao noted that CTA was delivering only about 84% of its promised service.

MIT developed a reinforcement-learning control strategy to improve bus headways. At a terminal, the system could recommend that a bus be held or released slightly earlier or later—typically by a few minutes—to maintain a more even spacing among buses.

Although the operational intervention is simple, determining the correct action requires accounting for significant uncertainty involving demand, available drivers, traffic conditions, incomplete information, and whether drivers will actually comply with the recommendation.



MIT's reinforcement-learning strategy for bus headway control, presented by Jinhua Zhao

The Importance of Local Knowledge

MIT initially used analytical criteria to recommend testing the system on a route with high ridership and low reliability. A CTA partner rejected the route because the depot involved would be unlikely to cooperate effectively with the experiment. That local organizational knowledge was not accounted for in the model. The team therefore chose another route whose supervisors and drivers were more willing to work with the researchers. Zhao used this story to illustrate what "human in the loop" actually means in practice: an AI system can be analytically correct while still selecting an operational approach that cannot work in the real organization.

The AI recommendation itself was also not sent directly to the bus. It went to a frontline supervisor who understood conditions at the depot and could decide how to communicate and implement the recommendation with the driver.

Results

MIT researchers were embedded throughout the CTA organization—from senior leadership to operating control centers, garages, supervisors, and drivers.

The experiment produced significant results:

- approximately 30% reduction in excess waiting time;
- 5–20% reduction in crowding; and
- improvement without adding buses or drivers.

The experiment also produced an important behavioral finding: drivers complied with AI recommendations only about 40% of the time, even in a depot selected partly because it was receptive to the pilot. Compliance fell when recommendations required larger departures from normal behavior, and drivers were more willing to delay slightly than to depart earlier.

For Zhao, this was an important example of why AI models need to incorporate human behavior rather than treating people as perfectly compliant components of a technical system.

Moving from Research to Agency Ownership

The first pilot relied heavily on MIT researchers embedded within CTA. In a subsequent phase, MIT stepped back and CTA employees themselves implemented the approach, with researchers serving only as advisors.

The objective was to ensure that knowledge and capability remained within the agency rather than creating a system that could operate only with continued academic involvement. Zhao said the approach was being expanded from Chicago to other major transit agencies.

"FANTASTICAL AI": EMERGING FRONTIERS

Zhao then shifted from practical implementation to several longer-term AI research frontiers.

Transportation World Models

Large language models have learned extensively from the virtual world but remain comparatively weak at understanding physical space and the interaction between physical infrastructure and human behavior. Zhao described the ambition of creating a transportation world model that understands both physical transportation systems and their underlying physics; and social systems, including human behavior. Such models could eventually strengthen digital twins by representing transportation as both a physical and human system.

AI for Transportation Science

Zhao also asked whether AI might eventually do more than analyze hypotheses supplied by researchers. Could AI itself generate genuinely useful hypotheses?

Historically, formulation of the research question has been regarded as a fundamentally human intellectual contribution. Zhao suggested that agentic systems may challenge that assumption, although humans would still need to judge which scientific directions matter and whether a technically interesting idea is valuable to society.

Multimodal Understanding of Cities

Different disciplines describe cities in fundamentally different ways: architects through images, anthropologists and historians through narratives, economists through numbers, and transportation analysts through networks.

Multimodal AI may offer a common analytical foundation capable of combining text, images, quantitative data, and network information, allowing researchers and planners to understand urban systems more holistically.

Recursive Self-Improvement and Multi-Agent AI

Zhao described experimental work in which AI agents not only propose solutions but also help define evaluation criteria, modify their own search processes, and share information through a persistent, shared memory.

The MIT research he described moves from a single agent toward multiple agents exploring problems in parallel and learning from each other's findings. In one experimental computer-science application, performance eventually exceeded the previous best-known result without researchers supplying the substantive solution—the humans established the architecture in which the agents could evolve.

Zhao acknowledged that this is both exciting and potentially unsettling. His concluding contrast was that while researchers should explore the frontier, transportation implementation should remain grounded in local conditions, real human behavior, organizational constraints, and development of in-house capabilities.

Donovan Guin — US Public Transportation Lead, IBM

ITS Optimization & Connected Vehicles

Guin, IBM's national lead for public transportation across state and local government, focused on the systems architecture required to realize the benefits described by the other speakers.

His core observation was that many transportation systems are already intelligent, but their intelligence is confined to individual silos.

A signal may optimize one intersection. A toll road may optimize traffic on that facility. A traffic management center may manage incidents within its own system. But those decisions may not account for consequences on adjoining roadways, transit systems, neighborhoods, or the rest of the network.

"NO AI WITHOUT IA"

Guin argued that large-scale AI depends first on information architecture (IA).

Rather than replacing every legacy ITS application, agencies can create an integration layer that allows meaningful information from different systems to be accessed together. A future statewide digital twin, for example, would not necessarily need every piece of raw data continuously. It would need appropriately structured, geocoded, time-referenced, governed information that can be combined across systems.

Only then can AI make inferences that span traffic, signals, connected vehicles, traveler information, and other functions.

FROM PREDICTION TO COORDINATED RESPONSE

Guin illustrated the concept using a potential congestion event.

One source—such as probe-vehicle data—might predict congestion 15 minutes in advance. But the operational question then becomes: what should the agency do?

In today's environment, people may have to move information manually from one system to another. Guin described traffic-management "war rooms" where employees look at different screens and verbally coordinate what each system is showing.

An agentic future could automate much of that coordination while retaining whatever level of human approval the agency chooses. One agent could detect a potential congestion event. Another could validate that conclusion using camera-based computer vision. A simulation could test response scenarios. Other systems could then modify traffic signal timing; send information directly to connected vehicles; generate traveler-information messages; and manage traffic flows

upstream and downstream of the event. The value comes from these systems operating together rather than from any single isolated AI application.

CONNECTED VEHICLES AND TRAVELER COMMUNICATION

Guin described opportunities for transportation agencies to communicate directly with vehicles through connected-vehicle platforms rather than requiring aftermarket equipment. He cited current work with North Carolina's turnpike system and examples involving manufacturer-integrated applications. The broader objective is a seamless exchange in which agencies can receive transportation information and then communicate relevant warnings or operational guidance back to travelers.

His overall message was that truly meaningful optimization requires integration, interoperability, connected vehicles, analytics, and AI working together according to a broader blueprint rather than a collection of isolated technology projects.

Discussion and Q&A

WHICH AI APPLICATIONS ARE PRODUCING THE HIGHEST ROI?

Chace asked the panel where agencies are seeing—or could see—the strongest return on investment.

Guin/IBM: Guin discussed congestion pricing and IBM's work with the 407 toll road in Toronto. Rather than treating each road user simply as a toll account, the system used customer and trip information to identify traveler segments and target incentives encouraging people to shift when they traveled. For example, a traveler might receive a discount for using the road at 10:00 rather than 9:00. Guin reported that while overall traffic continued to grow at approximately its expected rate, both peak congestion and crashes declined by about 5%.

Zhao/MIT: Zhao compared the current AI era with the earlier “productivity paradox” associated with computerization: enormous technology investment was visible everywhere before productivity gains appeared clearly in economy-wide statistics. He described MIT research on roughly 2,000 publicly listed companies that compared the proportion of AI talent in companies with measures such as productivity, ROI, and market capitalization. According to Zhao, greater AI staffing had not yet shown a significant relationship with most measures of actual productivity or ROI, although it was associated with higher market valuations. His interpretation was that AI may currently be producing clear micro-level improvements while broader organizational and economic productivity gains remain in an early “J-curve” stage.

Dwyer/AASHTO: Dwyer returned to the challenge of explaining major ITS expenditures to governors, legislators, and the public. Once technology investments become very large, they compete directly with bridge rehabilitation, pavement, and other visible transportation improvements. The business case therefore needs to quantify not only agency efficiency but reduced delay, freight benefits, safety benefits, and other user impacts.

HOW CAN AGENTIC AI BE ALIGNED WITH HUMAN VALUES AND PROTECTED AGAINST SECURITY RISKS?

An audience member asked Zhao how increasingly autonomous multi-agent systems can be constrained by moral and human principles and protected against hacking.

Zhao characterized this as a fundamental unresolved challenge. He described a capability-versus-alignment mismatch: enormous investment and talent are flowing toward making AI more powerful, while substantially fewer resources are devoted to ensuring that AI values and behaviors align with human values.

The alignment problem is difficult in part because humans themselves do not agree on a single set of values. Values differ across societies and cultures, and questions about what "purpose" should be embedded in AI rapidly become philosophical rather than merely technical.

He also noted the deeper question of whether humans can ultimately control systems that become more capable than humans themselves. His conclusion was not that the problem has been solved, but rather that alignment deserves far more attention alongside capability development.

Guin offered a different perspective based on technology economics. The enormous computing, electricity, cooling, water, and infrastructure requirements of current AI impose practical constraints. He suggested that smaller, fit-for-purpose language models may ultimately offer advantages for transportation because they can be more energy efficient, domain specific, and potentially easier to trust and govern.

WHAT IS THE NEAR-TERM FRONTIER FOR AGENTIC AI IN TRANSPORTATION?

Chace asked what transportation agencies should watch over the next six to twelve months. She noted that some early agentic systems are already finding value in applications such as responding to 511 customer inquiries and converting ATMS information into traveler information messages.

Guin/IBM: Guin pointed to New York City's interest in linking AI to its 311 intake systems for issues such as potholes and streetlights. He also discussed using dash-camera feeds from vehicles such as ride-hailing and delivery fleets to automatically identify infrastructure conditions. The technology itself may be deployable quickly, but public-sector procurement cycles remain a major constraint. He acknowledged that those processes can be frustrating but also protect agencies from making costly or risky decisions too rapidly. Pilots and research programs can therefore provide a useful bridge between technology development and formal procurement.

Zhao/MIT: Zhao encouraged transportation professionals to experiment with contemporary AI tools because the barriers to useful application have fallen dramatically. He observed that managing agentic AI resembles managing a human organization - designing communication among several AI agents is similar to deciding when an organization needs an all-hands meeting, a smaller team meeting, or more specialized communication. This suggests that leadership and organizational-design skills may become directly relevant to managing agentic AI.

Dwyer/AASHTO: Dwyer noted that state DOTs are already exploring more advanced AI agents for customer service, including responding to DMV questions. He shared an example of an agency testing such a system not by using subject-matter experts but by bringing in interns and asking them to try to "break" it. According to his account, the interns found weaknesses within two days. The example reinforced a recurring symposium theme: AI applications need adversarial, real-user testing—not simply confirmation by the people who built or understand the system.

Making AI Real: From Concepts to Practical Transportation Applications

Case Study Showcase 1

MODERATOR **Patricia Hendren, Executive Director, The Eastern Transportation Coalition**

SESSION DESCRIPTION

This session focused on how transportation organizations can move AI from interesting ideas and pilots into practical, sustainable use. Hendron challenged the panel and audience to consider: How would this actually work in my agency, organization, or company?

The panel offered three complementary perspectives:

- **Jennifer Volkening, Director of Data Analytics and Governance at Utah DOT**, described UDOT's first internally developed AI knowledge-search application and, importantly, the data-governance and organizational-adoption lessons learned from deploying it.
- **Thomas Gruenstaeudl from ASFINAG in Austria** described AI applications ranging from worker-safety and traffic prediction pilots to mature enterprise knowledge assistants named **Agnes** and **Adam**.
- **Jinhua Zhao of MIT** provided case studies from Washington, D.C., Singapore, and Chicago and used them to explore how AI can make customer intelligence actionable, accelerate planning and design, and change the roles of transportation professionals.

The discussion ultimately reinforced that successful AI implementation is less about selecting the newest model than about identifying an organizational need, preparing and governing the underlying information, building trust, experimenting, developing staff capability, and creating a pathway from a successful prototype to widespread adoption.

SESSION TAKEAWAYS

- **Start with the problem, not the AI.** A recurring theme was to begin by asking "why"—what problem is consuming staff time, frustrating customers, creating safety risks, or limiting agency performance—and only then determine whether AI is the appropriate tool. AI technologies change on a timescale of weeks, while transportation services change much more slowly and infrastructure over decades; strategies built around a particular technology can therefore become obsolete almost immediately.
- **A simple, low-risk internal use case can be an excellent first AI project.** UDOT began with searching internal policies and procedures. The tool did not perform a safety-critical transportation function; it solved an everyday employee problem and generated immediate, visible time savings.
- **AI quickly exposes weaknesses in information governance.** UDOT discovered that incomplete document collections and multiple versions of the same policy caused incorrect or inconsistent answers. The AI project therefore became a data- and knowledge-governance project as much as a technology project.
- **The harder knowledge-management problem is increasingly retrieval, not creation.** ASFINAG's experience was that the agency already had enormous amounts of information—in policies, manuals, SharePoint sites, file shares, correspondence, and specialized systems. The challenge was finding the right information at the moment it was needed.

- **Privacy and information security should shape architecture from the beginning.** ASFINAG needed to comply with European GDPR requirements, leading it toward building internally controlled enterprise AI environments rather than using commercially available AI services.
- **Make unstructured information actionable, not merely searchable.** Zhao's Washington Metropolitan Area Transit Authority example went beyond detecting positive or negative sentiment. AI categorized customer comments by topic, route, place, time, and responsible business unit so that someone could actually take action.
- **AI may dramatically shorten the feedback cycle in transportation planning.** A resident suggestion that once might require months of consultant analysis could eventually be explored and visualized during the engagement process itself, allowing more iterative public participation.
- **Technology development may be faster than organizational adoption.** Zhao distinguished the "0-to-1" challenge of developing a working method from the much longer "1-to-100" challenge of institutionalizing it across an organization. A London application took roughly three years to develop and another four years to spread across multiple departments.
- **Measure augmentation as well as automation.** Automation metrics answer how much work AI replaces. Augmentation asks whether people can perform work that was previously beyond their capability. Zhao argued that agencies currently measure the former much better than the latter.
- **Accountability may become more valuable than raw production.** AI can produce memos, code, reports, and correspondence rapidly, but a qualified person must still verify the output and accept responsibility for it. Professional expertise increasingly resides in knowing whether the answer is defensible.
- **Adoption requires deliberately engaging skeptics, not just enthusiasts.** UDOT's statewide AI roadshow included both a "dream board" for possible applications and a "Therapy Corner" where employees could voice concerns about trust, plagiarism, intellectual property, and other issues.
- **Data ownership, intellectual property, and organizational knowledge are strategic AI issues.** The closing discussion raised a deeper concern: when proprietary organizational knowledge is fed into an external model, who ultimately owns and benefits from that accumulated knowledge?

Jennifer Volkening — Director of Data Analytics and Governance, Utah DOT

Volkening described UDOT's initial AI implementation project to create a tool that allows employees to ask questions about agency policies and procedures and receive the information they needed without trying to determine where documents are located and manually searching through them.

Usage grew gradually, and employees began telling Volkening that the application was saving them hours of searching through internal policies and procedures.

WHY POLICY SEARCH?

UDOT did not begin with a comprehensive AI strategy. The project originated partly from curiosity among the data analytics team. The team looked for something that consumed significant employee time; had an existing body of source material; could be relatively easily bounded; and could provide a consistent, standardized answer.

Searching internal policies fit those conditions well. This made the project particularly useful as an organizational learning exercise. It generated a practical benefit while allowing UDOT staff to learn how to build, test, govern, deploy, and maintain an AI application.

THE "BOOT REIMBURSEMENT" LESSON

One of the most useful lessons emerged during beta testing.

The initial model included policies but not procedures. When beta users were asked to test the system, one of the most common questions was essentially, "How do I get my boots reimbursed?"

The tool could not answer correctly—not because the AI was malfunctioning, but because the answer existed in a procedure**,** which had not been included in the source material.

UDOT subsequently added procedures to the knowledge base, and the system could then answer the question appropriately.

This illustrates an important implementation principle: When AI gives an incomplete answer, the problem may be the knowledge environment rather than the model.

DOCUMENT VERSION CONTROL

UDOT encountered an even more consequential governance problem when the system had access to two different versions of the same policy.

Because the model did not know which one was authoritative, it could return information from an outdated version. One example involved employee conduct or dress-code guidance.

The experience demonstrated that deploying AI over organizational documents requires authoritative source identification; document ownership; version control; retirement of superseded information; and continuing monitoring of source material.

KEY LESSON

The important lesson from UDOT was not merely "build a chatbot." It was that an apparently simple chatbot can expose underlying problems in how an organization manages its knowledge. Improving the AI may therefore require improving the institution's underlying information-management practices.

UDOT'S NEXT STAGE: BROADER ADOPTION

During Q&A, Volkening described an agency discovery tour designed to understand both AI use and organizational attitudes. UDOT has approximately 1,800 employees and four regions. The team held six open houses at different locations, including locations intended to capture perspectives outside headquarters. Attendance at most locations was substantial relative to UDOT's size, often 50–60 people.

Volkening characterized UDOT's current adoption level as somewhere in the middle: many employees are curious and eager to experiment, but skeptics remain. Employees have expressed concerns related to trust; transparency; plagiarism; appropriation of other people's work or ideas; and appropriate AI tools and usage. Rather than dismissing these concerns, UDOT explicitly invited them into the conversation.

Thomas Gruenstaeudl — Head of Project Development, ASFINAG GmbH, Austria

Artificial Intelligence in the Road & Mobility Sector: Operational AI Solutions & AI Pilot Projects

Gruenstaeudl provided a European perspective through ASFINAG, Austria's integrated motorway organization. ASFINAG builds, operates, maintains, tolls, and finances the motorway network and is funded through user toll revenues. He presented both transportation-operational AI pilots and enterprise knowledge-management applications.

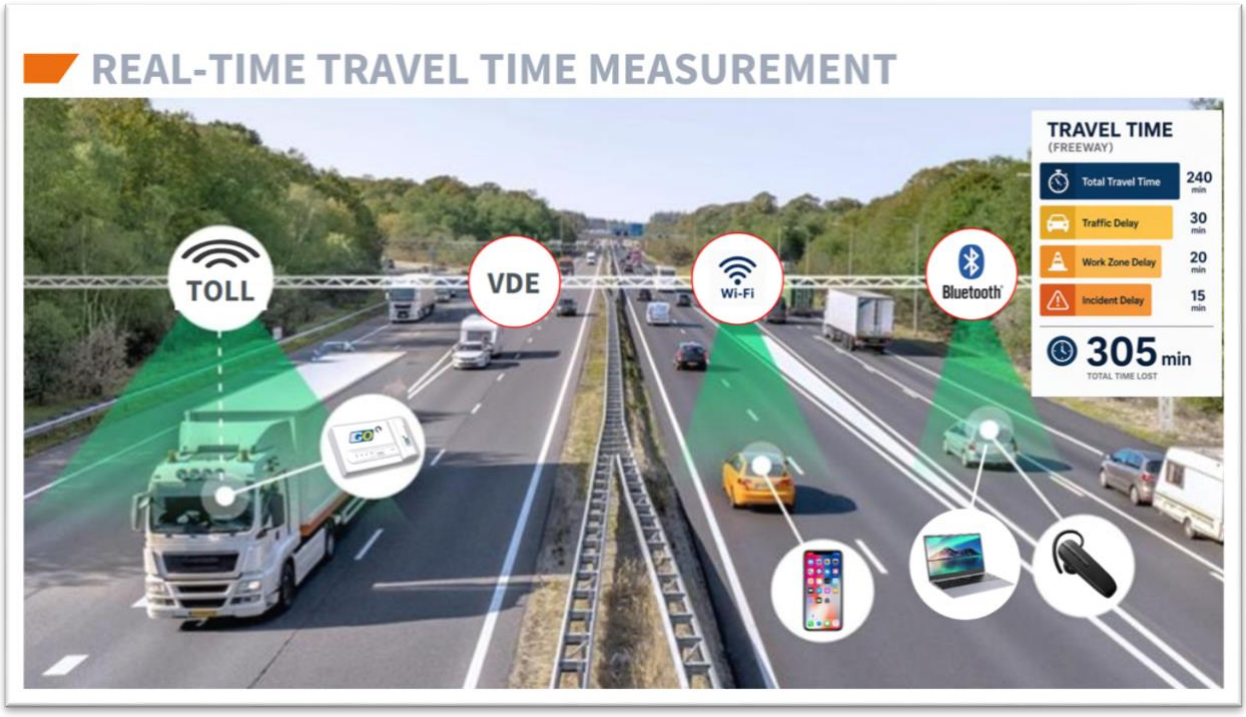
COLLISION EYE: PROTECTING MAINTENANCE WORKERS

ASFINAG is piloting an AI-based warning system for road maintenance workers called **Collision Eye**, which uses sensing and AI to warn workers about potentially dangerous approaching vehicles. The example illustrated a useful approach to selecting AI applications: begin with a specific, high-consequence problem where faster or better sensing can directly improve safety.

PREDICTIVE TRAVEL TIMES

ASFINAG also operates a large information environment that includes traffic sensors; tolling information; more than 1,100 cameras; historical traffic patterns; roadwork and lane-availability information; weather forecasts; major events; and real-time traffic information.

AI is being tested to integrate these sources and forecast traffic conditions not only one hour ahead, but four hours and potentially as much as a week in advance. Instead of simply telling customers what congestion exists now, the organization wants to anticipate what conditions will be and allow both the agency and travelers to respond earlier.



ASFINAG's Predictive Travel Time Pilot, presented by Thomas Gruenstaeudl .

AGNES: ENTERPRISE KNOWLEDGE ASSISTANT

ASFINAG's more mature AI experience began with information retrieval. Like many large organizations, ASFINAG's institutional knowledge is distributed across policies, handbooks, SharePoint sites, file shares, databases, and other locations.

Public generative AI tools could not simply be used for this material because ASFINAG information could be confidential; and the organization must comply with the European Union's General Data Protection Regulation (GDPR). That led ASFINAG to create an internally controlled enterprise knowledge assistant called Agnes.

Employees can ask a natural-language question such as how to submit an expense report and receive a description of the process, a copy of the appropriate form, and a reference to the applicable guideline or manual chapter. Rather than searching multiple systems or asking colleagues, employees can go directly to Agnes.

Gruenstaeudl reported:

- roughly 1,700 registered users;
- about 600 active users; and
- more than 50,000 questions answered, with user feedback collected through thumbs-up/thumbs-down responses.

ADAM: PERSONALIZED OR TEAM-SPECIFIC KNOWLEDGE ASSISTANTS

Whereas **Agnes** addresses enterprise-wide knowledge, **Adam** allows departments and employees to build more specialized assistants around information relevant to a particular function.

Gruenstaeudl provided an example of how one work group used Adam to create a specialized chatbot that searches historical records including old project files, correspondence, and historical commitments. This chatbot was used to research a customer inquiry about the status of a noise barrier that was promised 5 years ago. It allowed an employee of the work group to reconstruct the history in seconds rather than searching manually for hours.

USAGE AND ECONOMICS

During Q&A, Gruenstaeudl reported that Adam had approximately 300 users but more than 200,000 requests annually, apparently exceeding Agnes in total request volume. He said each request cost roughly three cents, and his IT colleagues estimated that providing premium Microsoft Copilot licenses to everyone would currently cost perhaps 20–30 times as much.

MOVING TOWARD AGENTIC PROCESSES

An audience member asked whether Agnes and Adam could eventually move beyond telling someone where a form is located and actually perform parts of the process.

Gruenstaeudl said that is the intended direction but estimated the organization is still perhaps one to two years away from substantial process-level automation.

One specific next application is procurement. ASFINAG conducts approximately 1,500–2,000 procurement procedures per year and must work with roughly 22,000 Austrian standards. Thomas sees significant potential for AI to combine applicable standards with ASFINAG's internal rules to make preparation of tender documents faster and more consistent.

ASFINAG'S CORE LESSON: TRUST AND ADOPTION

Gruenstaeudl closed his presentation by emphasizing that the largest challenge is not purely "artificial intelligence" but human intelligence, trust, and adoption.

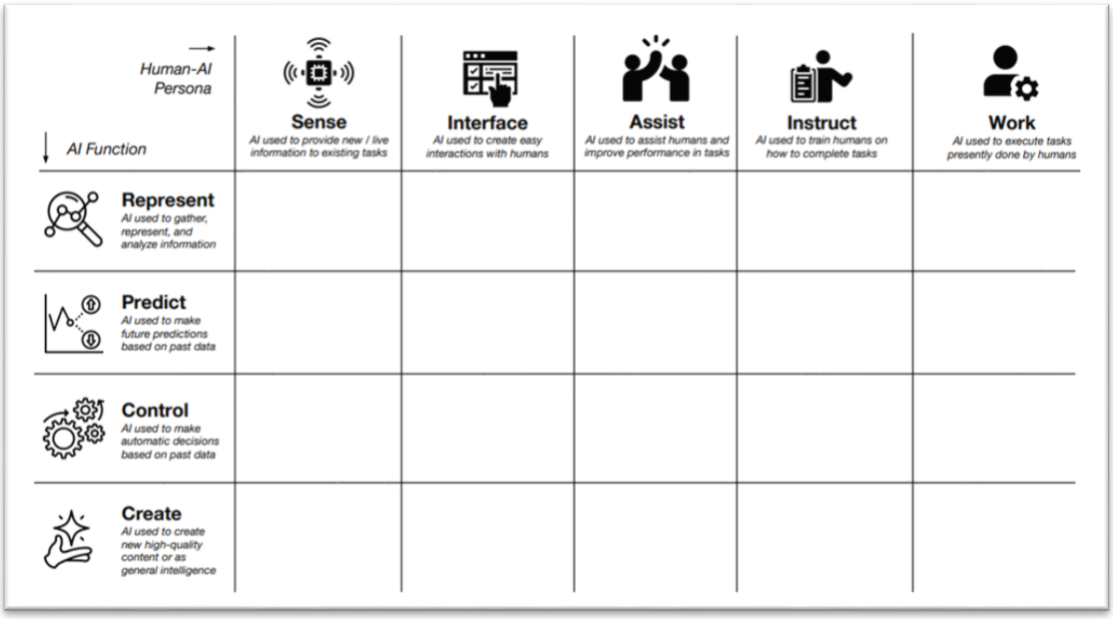


ASFINAG's Employee Training Strategy, presented by Thomas Gruenstaeudl .

That observation aligned closely with UDOT's experience: the underlying technology can be relatively straightforward compared with building an authoritative knowledge environment and getting employees to trust and incorporate the tool into daily work.

Jinhua Zhao — Professor of Cities and Transportation, MIT

Zhao used the session to complement his earlier presentation with additional **real-world and near-real-world case studies**. He provided a useful framework for categorizing AI applications, shown below:



A framework for categorizing AI applications in transportation, presented by Jinhua Zhao.

WMATA - AI FOR CUSTOMER INTELLIGENCE AND SENTIMENT ANALYSIS.

A model was created for the Washington Metropolitan Transportation Area Authority (WMATA) to automate categorization of social media posts, and route specific complaints to appropriate business units. The data stream consisted of approximately 100,000 posts per day. Posts were tagged based on transportation mode; route, location, time, issue, topic, sentiment, and level of specificity. Using AI to process social media posts provides an alternative to periodic customer surveys, allowing an agency to monitor customer concerns over time, map where problems occur, assess whether interventions affect sentiment; determine whether a policy change is working; and potentially generate a daily picture of customer experience. It can turn customer sentiment into something executable and actionable, rather than merely measuring satisfaction.

CHICAGO - AI FOR GENERATIVE URBAN DESIGN.

MIT applied ControlNet (a neural network architecture used to add spatial control to AI image generators) to demonstrate an application for a quick response iterative urban design process. The research used generative models trained to associate land-use descriptions with spatial representations. Once trained, the system could generate alternative plans within minutes while observing specified physical constraints, land use targets, and desired urban textures.

SINGAPORE – MULTIPLE APPLICATIONS

Zhao also discussed examples from Singapore, where MIT has worked on transportation technology for roughly a decade. He characterized Singapore's approach as systematic across planning, forecasting, traffic management, passenger experience, crowd management, operations, and data platforms. While transferability may be limited give that Singapore's political, institutional, cultural, and governmental context differs substantially from the United States, he identified several notable examples including **Glide**, an AI adaptive signal control application; a bus-load forecasting application, and a unified transportation operations platform.

He drew several implementation lessons from the Singapore experience:

- build a unified data platform;

- reduce barriers to legitimate data sharing between government agencies;
- deliberately break down departmental silos;
- leverage existing sensors, cameras, and systems before purchasing entirely new infrastructure;
- demonstrate value quickly; and
- make appropriate data available externally so researchers and others can develop additional value from it.

He also pointed to two important aspects of Singapore's approach to technology deployment:

- A willingness to experiment early, learn, and iterate with emerging technology
- Approaching technology deployment with questions about how technology could be used to improve rather than replace existing capabilities (e.g., instead of "can an autonomous vehicle replicate what today's driver does", ask "how could autonomous vehicles change and improve urban management itself?)

IMPLICATIONS

Zhao concluded his remarks with some implications for transportation agencies and their workforces:

- **Immediacy:** Transportation professionals will be expected to respond faster
- **Accountability:** AI may generate a report, plan, email, policy memo, or design, but someone still needs to validate it and accept responsibility for the result
- **Augmentation:** AI will allow staff to accomplish something they could not previously accomplish, or to operate at a much higher level. (This type of impact needs to be considered within ROI analysis, but is not straightforward to measure)

Discussion and Q&A

WHAT IS THE ROI?

An audience member returned to a question from the previous session about ROI and asked Gruenstaeudl whether ASFINAG had calculated it.

Gruenstaeudl said no formal economic analysis was performed at the outset. The organization began experimenting and learned from actual deployment. Now, usage statistics provide a practical basis for understanding value. Adam reportedly processes more than 200,000 requests annually at approximately three cents each, while broad deployment of premium Microsoft Copilot licenses was estimated to cost substantially more under current usage patterns.

Hendren observed that public agencies will continually be expected to defend AI investments and therefore need to develop stronger evidence on return.

Another participant noted that economics could change as usage grows because one product might charge per request while another uses a fixed per-user license. Gruenstaeudl agreed that ASFINAG will have to continue reevaluating the economics as technology and pricing evolve.

HOW FAR CAN KNOWLEDGE ASSISTANTS GO TOWARD AGENTIC AI?

An audience member asked whether systems such as Agnes could eventually stop merely directing someone to a form or process and instead execute steps in that process.

Gruenstaeudl agreed that this is a logical future direction but said ASFINAG remains perhaps one to two years from that stage. Procurement is a particularly attractive next application because the

organization conducts thousands of tender processes and must reconcile internal rules with tens of thousands of standards.

WHAT DETERMINES ADOPTION AND LONGEVITY?

An audience member asked all three panelists how widely their tools had been adopted and whether they would remain useful given how rapidly AI is changing.

Volkening/UDOT: Volkening described adoption as mixed but encouraging. Many employees are highly interested, and the statewide discovery tour drew significant attendance, but skeptics remain. UDOT's response has been to actively engage both groups and collect ideas and concerns rather than treating adoption as simply a communications campaign.

Zhao/MIT: Zhao said organizational diffusion typically takes much longer than developing the technical innovation. He described this as the difference between “0 to 1” innovation and “1 to 100” adoption. A successful pilot is not the end of implementation; in many ways it is only the beginning.

WHO OWNS THE DATA, THE AI OUTPUT, AND THE KNOWLEDGE?

The final audience question concerned ownership of underlying data, AI-generated information, and resulting intellectual property.

Zhao considered this a major strategic issue. He noted that contractual protections and enterprise “sandbox” environments help to protect proprietary information, but organizations still face questions about how much they can trust vendor commitments and how effectively legal remedies can protect knowledge after it has escaped.

This is one reason organizations are increasingly interested in locally deployable or open-source models, where they have substantially greater physical and technical control over information.

The broader copyright question—who owns the enormous body of human-created knowledge used to train foundational AI models—remains unresolved.

DAY 1 · SESSION 07

Building an AI Strategy: Governance, Workforce, Implementation, and Organizational Readiness

Peer Roundtable 1

MODERATOR **Craig Thor, Chief Scientist, FHWA Turner-Fairbank Highway Research Center**

SESSION DESCRIPTION

This session focused on how transportation agencies and related organizations can build an AI strategy that translates rapidly evolving technology into practical, sustainable organizational results. Thor framed the discussion around a recurring theme from the symposium: the technology itself is advancing quickly, but many of the hardest challenges involve implementation, governance, organizational readiness, workforce development, data, and change management. He emphasized that traditional strategic planning approaches may need to become more flexible because AI changes much faster than most transportation organizations. An AI strategy must consider not only what tools to deploy, but also who agencies should hire, how existing

employees should be upskilled, what data and governance structures are required, and how to ensure that technological innovation produces actual transportation outcomes.

The four panelists brought different perspectives:

- **Alex Walcher of ASFINAG and PIARC** discussed how the World Road Association is using AI to help develop its next strategic plan across 128 member countries.
- **Giri Venkiteela, Innovation Officer at New Jersey DOT** described NJDOT's evolving AI journey, emphasizing literacy, small pilots, accountability, leadership support, and peer learning.
- **Robert Sheean, FHWA** discussed FHWA's effort to create a practical national AI roadmap and resource environment that agencies can use across the full range of DOT functions.
- **Mike Snodgrass, Applied AI Specialist at Google**, provided a private-sector perspective emphasizing use cases, measurable ROI, internal capability, collaboration, and targeted AI deployment rather than technology for technology's sake.

The session then expanded into a substantial discussion of workforce, education, data, security, metrics, change management, and how public agencies can maintain momentum when technology is moving much faster than their traditional institutional processes.

SESSION TAKEAWAYS

- **An AI strategy should begin with organizational goals and problems, not with AI itself.** Multiple speakers cautioned against implementing AI simply because the technology is available. Agencies should define a specific problem, workflow, desired outcome, and accountable owner before selecting the technology. NJDOT summarized this simply: "you cannot implement AI for the sake of AI."
- **An AI strategy must be flexible.** Transportation agencies work with planning cycles, budgets, procurement processes, and infrastructure that change over years or decades, while AI capabilities can change within weeks or months. Strategic plans therefore need durable objectives but adaptable technology choices.
- **AI can support strategy development, but should not make strategic choices.** AI can surface options and manage complexity, but people must determine the strategy.
- **AI literacy is an organizational capability, not just an IT skill.** Transportation engineers, planners, managers, executives, and frontline employees do not need to become data scientists, but they do need to understand what AI can and cannot do, what hallucination means, how models use data, and where professional judgment remains essential.
- **The future transportation workforce needs both domain expertise and AI literacy.** The panel did not conclude that DOTs should replace engineers with data scientists. Instead, it pointed toward professionals who understand transportation deeply while also knowing how to use, question, and govern AI.
- **Pilot small, learn, document, then scale.** Even a simple project can build confidence if the agency clearly defines the problem, prepares the data, tests and validates the application, identifies limitations, evaluates results, and documents lessons before scaling.
- **Assign accountability from the beginning.** AI governance cannot be separated from the question: Who is accountable if the application is wrong or fails?
- **Leadership support and grassroots experimentation need to coexist.** Employees often have more enthusiasm for AI than leaders expect, but executive support is needed for funding, governance, sanctioned tools, and scaling successful applications.
- **Measure business outcomes, not AI activity.** Examples included reducing a 21-day process to two days, eliminating a large share of a permitting backlog, improving quality in less time, or measuring how AI affects service delivery.

- **AI is increasingly exposing existing data problems.** Agencies often discover only after beginning an AI application that necessary datasets cannot be shared, are poorly governed, or remain trapped in organizational silos. That can be productive: a practical use case can reveal exactly which data problems need to be fixed.
- **Protect the development of professional judgment.** One important concern was that young engineers traditionally acquire judgment by performing tasks, seeing results, and learning from experience. If AI performs too much of that early-career work, agencies and universities must intentionally ensure that future professionals still develop the expertise needed to evaluate AI outputs.
- **Peer learning may be one of the fastest paths to responsible adoption.** DOTs repeatedly face similar functions, data, processes, and institutional barriers. Rather than independently reinventing solutions, the panel encouraged sharing pilots, lessons, governance practices, and applications across states and federal partners.
- **Change management may ultimately be as important as the technology.** The closing discussion emphasized that public agencies often do not dedicate enough resources to organizational change, yet AI adoption will require changes in jobs, responsibilities, workflows, training, leadership expectations, and culture.

Alex Walcher — Managing Director, ASFINAG; Vice President, PIARC

Walcher provided an overview of PIARC – the World Road Association, founded in 1909. PIARC brings together experts from road administrations across 128 member countries to share knowledge about roads and road TRANSPORT. PIARC's members represent highly diverse economic, geographic, institutional, and transportation contexts. Its technical committees operate in four-year cycles, and every four years PIARC develops a strategic plan that establishes both overall direction and the technical work program for the next cycle.

Walcher focused his remarks on how PIARC is using AI to create its broader organizational strategy. Traditionally, that process requires extensive consultation with technical committees, delegates from member countries, and experts and working groups. The result is an enormous and complex body of information that must be reconciled for gaps, conflicts, overlaps, and priorities. The traditional consultation process can take approximately a year and involve numerous meetings and iterative reviews. Walcher proposed using AI to synthesize information and accelerate preparation of the strategic framework.

Two years earlier, when he first suggested this approach, members of PIARC's Strategic Planning Commission were concerned that AI was too risky and that they did not understand what would happen to the information. By the time of the symposium, using AI in the process had become largely accepted. Walcher used this rapid shift in attitude to illustrate how quickly institutional expectations around AI are changing. It was important for people to understand that AI can greatly improve the analytical and synthesis portions of strategic planning, but purpose, values, priorities, and final decisions remain human responsibilities.

Giri Venkiteela — Innovation Officer, New Jersey Department of Transportation

From Questions to Action: NJDOT's Journey

Venkiteela described NJDOT's AI journey as an evolving institutional process rather than a finished strategy. His presentation emphasized the need to bridge the gap between AI researchers and transportation engineers.

New Jersey established a statewide AI policy in 2023 and provided state employees with an approved generative AI assistant.

NJDOT's effort began with a research and innovation group asking fundamental questions:

- Where could AI actually help DOT functions?
- What could it do beyond drafting emails and summarizing documents?
- Could it assist bridge work, payments, procurement, or other operational processes?
- What risks and governance structures would be necessary?

NJDOT then participated in peer exchanges with other states and found that most agencies were asking the same questions. By 2025–2026, NJDOT had moved further into educating employees, identifying use cases, conducting pilots, and collaborating nationally.

WORKFORCE EDUCATION

Venkiteela said employee education had become one of the agency's most important immediate priorities. For NJDOT, AI literacy is not synonymous with teaching people how to use ChatGPT. Employees should understand at least at a conceptual level what an AI model is, what goes into it, what it can and cannot do, why hallucinations occur, why validation is required, and the importance of a human taking responsibility for the results.

NJDOT has used activities such as lunch-and-learns, cross-department workshops, and executive briefings to build that understanding.



Building AI understanding across the agency, presented by Giri Venkiteela, NJDOT.

An interesting finding was
that staff were often more enthusiastic than expected. Rather than resisting AI, some employees wanted to apply it to almost everything. The agency's challenge was therefore not merely encouraging adoption but helping employees distinguish appropriate from inappropriate applications.

PILOTS

NJDOT deliberately encourages small pilots. A project does not have to be transformational to be worthwhile. Small applications help employees become comfortable with AI and give the organization practical experience with activities such as data preparation, testing, validation, documentation and maintenance. The objective is to learn before scaling.

Venkiteela strongly cautioned against implementing AI "for the sake of AI." A pilot should have a clearly defined problem and scope, appropriate source data, known risks, a designated team with access to required subject-matter and AI/development expertise, and participation from IT.

NJDOT specifically assigns accountability among subject-matter experts, an AI liaison, developers, and IT partners. Venkiteela noted that agencies often focus heavily on developing an AI system but much less on what happens afterward.

For every pilot, NJDOT wants to capture information about what was tested, how well it worked, ROI (where measurable), employee training involved, maintenance requirements, and ongoing model retraining needs. A model that performs well today may not remain appropriate tomorrow, so maintenance must be considered part of AI strategy rather than an afterthought.

MULTI-STATE COLLABORATION

NJDOT is the lead state for the FHWA-supported pooled fund effort TPF-5(573) - State DOT Council for Strategic AI Adoption intended to provide an AI implementation network for State DOTs, ensuring they have the capacity, resources, and peer support to translate national AI strategies into operational deployments. DOT organizational structures and functions have many similarities, so agencies should not continually reinvent the same solutions independently.

Robert Sheehan — Director, FHWA AI Deployment Center of Excellence

Artificial Intelligence Deployment Center of Excellence

Sheehan focused on the federal role in helping transportation agencies convert scattered lessons and pilots into practical resources that can be used across the entire DOT community.

He described his role not as providing all the answers himself, but as creating an environment where lessons are collected, documented, organized, made accessible, translated into usable tools, and continually updated. This perspective helped motivate the pooled-fund study, an AI roadmap, and a national resource portal.

AI MUST ADDRESS THE WHOLE DOT

Based on recent peer exchanges, agency interest in AI is quite broad, spanning executive management, administration, and core business functions (planning, project development and delivery, multimodal coordination, and maintenance and operations)

The roadmap therefore needs to focus not merely on individual technologies but on how AI fits into DOT business processes and workflows across the organization.

FEDERAL RESEARCH MUST BECOME MORE AGILE

A significant institutional challenge is that AI evolves far faster than traditional public-sector research and procurement cycles.

Sheehan pointed to the difficulty of responding rapidly through fiscal-year funding and acquisition processes. A technology area that changes within months cannot be supported effectively if research initiation, contracting, and funding take years. There is therefore a need for mechanisms that support rapid testing of concepts, quick dissemination of findings, sharing of prototypes across agencies, and ongoing conversations between formal conferences and research cycles.

ALWAYS ASK "WHY?"

Sheehan returned repeatedly to a lesson from an introductory engineering assignment: before building a better mousetrap, ask why you are building it. The same applies to AI. The objective is not to advocate AI everywhere. It is to provide a framework for looking at what work is being performed; why AI might improve it; what outcome is expected; what safety or trust implications exist; and whether AI is actually the right approach.

The federal focus is therefore on safe and trustworthy AI, not universal AI implementation.

WORKFORCE AND ENGINEERING JUDGMENT

Sheehan raised one of the session's most significant workforce questions.

Young engineers traditionally develop professional judgment through years of practice. If AI automates many entry-level analytical activities, what experiences will future engineers have from which to develop judgment? That concern is not simply about whether jobs disappear. It is about whether professionals will still develop the experience required to recognize when an AI-generated answer is wrong. Sheehan suggested that future workforce strategy may require some blending of transportation domain knowledge and data/AI capability..

Mike Snodgrass — AI Specialist, Google

Snodgrass provided a private-sector view based on working with DOTs, higher-education institutions, and other government agencies. His central message was that AI has become accessible enough that organizations can increasingly develop applications themselves rather than depending entirely on external vendors**.** He contrasted AI with some earlier IT implementations. Employees can now experiment directly—for example, creating an internal policy-search application—and quickly understand whether it provides value.

FOCUS ON THE USE CASE, NOT THE MODEL

For many common business applications, Snodgrass argued that differences among leading models are less important than identifying the right use case. Agencies should target current bottlenecks – processes that are currently slow or frustrating that AI could potentially improve, produce measurable outcomes, and quickly demonstrate value. Examples might include reducing a permitting backlog by 75%; or reducing a 21-day process to two days. Those are the kinds of outcomes that make sense to governance committees and executives because they describe organizational results rather than AI capabilities.

EMBED AI INTO PERFORMANCE EXPECTATIONS

Snodgrass suggested that organizations can go beyond simply giving employees AI tools. AI-related process improvement could eventually appear in job descriptions; annual objectives; or

performance conversations. Managers might ask: How did you use AI to improve a process this year? This would move AI from optional experimentation into a normal element of organizational performance and continuous improvement.

COLLABORATION IS INCREASING

Snodgrass noted that he has seen much greater cross-agency collaboration than in many previous technology areas. DOTs are increasingly looking for opportunities to build on what their peers have done. He saw substantial opportunity for shared learning and reduced duplication.

ADDRESS DATA PROBLEMS ITERATIVELY

Snodgrass observed that AI pilots often expose problems that agencies have had for years. For example, an agency may build a promising application only to discover that the required datasets have unclear ownership, lack consistent standards, or cannot legally or organizationally be shared.

Rather than employing a comprehensive data cleanup approach, Snodgrass suggested allowing useful applications to reveal the specific data barriers that matter and then addressing those barriers iteratively.

AI AS A THOUGHT PARTNER, NOT A SUBSTITUTE FOR THINKING

On workforce questions, Snodgrass acknowledged concern that people may “offload” too much thinking to AI. His view was that the strongest employees will likely be those who can use AI as a thought partner to increase their effectiveness rather than simply treating it as a replacement for expertise. He repeated a commonly heard observation: AI itself may not replace a professional, but someone who knows how to use AI effectively may have an advantage over someone who does not.

Discussion and Q&A

WHAT SHOULD AI DO—AND WHAT SHOULD HUMANS STILL DO?

Thor asked how organizations should decide where AI belongs in a strategic plan given that AI is not appropriate for everything.

Walcher/PIARC/ASFINAG: AI should be viewed as a tool to manage complexity, synthesize large amounts of information, surface options, and help people think. Humans must evaluate the alternatives and decide which path aligns with the organization's mission, priorities, values, and responsibilities.

Sheehan/FHWA: Begin with the work and intended outcome rather than assuming an AI application is needed.

HOW SHOULD AGENCIES MEASURE SUCCESS?

The discussion highlighted several levels of measurement.

Snodgrass/Google: Measure concrete business outcomes such as backlog reduction and cycle-time improvement.

Walcher/PIARC/ASFINAG: Compare both task completion time and quality before and after introduction of AI. Keep in mind that AI can occasionally make work more complicated, meaning agencies should be willing to conclude that a particular application is not providing sufficient value.

WHAT HAPPENS TO THE WORKFORCE?

Thor asked the panel to comment on workforce implications of AI – in terms of needed skills, changes to job classifications, how employees should interact with AI, and what the workforce should look like five to ten years from now.

Sheehan/FHWA: There are real concerns about how future engineers acquire experience if AI performs many traditional early-career tasks. Professional judgment remains essential because someone still needs to determine whether a recommendation is appropriate in the real transportation environment.

Snodgrass/Google: Workers who can use AI effectively as an analytical and creative partner may become more valuable.

Walcher/PIARC/ASFINAG: Just as with previous technological transformations, some tasks and jobs may disappear, new ones will emerge, and workers and organizations will have to adapt. Ignoring AI could ultimately be more dangerous than engaging with it, particularly for younger generations.

Venkiteela/NJDOT: Today's younger employees often enter the workforce already expecting to use AI and modern digital tools. If DOTs do not provide those opportunities, agencies may lose talent—not because AI replaces workers, but because workers leave for organizations where they can use contemporary technology.

WHAT SHOULD UNIVERSITIES TEACH?

An audience member asked what message the panel would give universities preparing the future transportation workforce.

Venkiteela/NJDOT: Teach the fundamentals of AI to civil engineering students. They should be aware of how AI can be applied to civil engineering problems and have a conceptual understanding of how models work; what prediction accuracy means; and what limitations exist. Hands-on experimentation with relatively simple AI models could build comfort and critical understanding.

Sheehan/FHWA: There is a need for cross-disciplinary education. AI students need exposure to real transportation applications, while engineering students need enough AI understanding to use the technology responsibly. Universities could create stronger connections between engineering, computer science, data science, planning, and other programs rather than treating them as isolated disciplines.

Snodgrass/Google: Education should encourage experimentation but also teach the “guardrails” around applications. Students need to be aware of privacy implications, and model biases and limitations. They need to be able to judge whether model outputs can be trusted. A well-rounded graduate should understand both what AI makes possible and what risks accompany it.

Walcher/PIARC. Students should understand what the technology can and cannot do and recognize that it supports rather than replaces professional judgment.

HOW SHOULD DATA AND SECURITY FIT INTO AI STRATEGY?

Snodgrass/Google: Agencies should use promising applications to identify the specific data barriers that matter most rather than trying to fix every enterprise data problem.

Sheehan/FHWA: Develop a data strategy linked to the AI strategy. AI may help work around some inconsistencies, but it does not eliminate the need for standards, governance, ownership, and trustworthy information.

DAY 1 · SESSION 08

Day 1 Reflections

Wrap Up

MODERATOR **Hyun-A Park, President, Spy Pond Partners**

SESSION DESCRIPTION

The first day of the symposium ended with reflections on the day's presentations. Park began by summarizing some key themes, and then asked participants to add their own thoughts.

SESSION TAKEAWAYS

- **AI has many diverse applications for transportation agencies.** It helps agencies to improve predictive capabilities, which enables them to be more proactive on multiple fronts including safety, operations, and asset management. It also helps agencies to automate repetitive tasks and reduce cycle time for improved efficiency. It helps agencies to improve service the public.
- **AI provides the tools to make the unobservable observable.** Examples include using AI to derive information about intersection safety based on camera feeds, finding patterns in social media data streams, processing bridge inspection imagery.
- **Successful AI deployment requires attention to people and process.** Key "infrastructure" for AI includes governance, data management, workforce training including AI and data literacy.
- **Agencies need to make the case for AI investment.** There is a need for effective communication about this to agency leaders – and in many states there is frequent turnover of leadership. ROI discussions need to cover efficiency, effectiveness, public service – and also address the question of "what is the cost of not adopting AI?"
- **There is a need for collaboration in fostering AI adoption.** Agencies need to learn from one another and reuse successful approaches (when appropriate) rather than reinventing the wheel. Many agencies are now conducting small AI pilots, and a credible track record elsewhere can make the next agency more comfortable experimenting. We heard from several organizations that are working to help with this process: USDOT, AASHTO, ITS America, TRB, PIARC . Further conversations about role clarity across these different organizations would be beneficial.
- **AI evolves at technology speed, while public agencies operate at institutional speed.** Strategic success will therefore depend less on predicting which AI model will dominate in several years and more on building organizations that are capable of learning, adapting, governing, and implementing new capabilities as they emerge.
- **Change management is critical.** Given the pace of change in AI technology and its implications for workforce development and business process adaptation, agencies need to improve their internal capabilities for change management. Without deliberate attention to organizational change, even technically excellent AI applications may never become widely used or sustained.

Day 2

Wednesday, August 20, 2026

01 Day 2 Keynote and Fireside Chat — Seval Oz, U.S. Department of Transportation
02 AI in Planning and Project Development
03 AI in Asset Management, Predictive Maintenance, and Inspection
04 From Data to Digital Twins: The Future of Asset Management
05 Data Fusion: Building the Data Foundation for AI-Enabled Transportation Decisions
06 Conference Takeaways and the Path Forward for AI in Transportation

DAY 2 · SESSION 01

Mobility 2040: AI's Role in Next-Gen Transport — Seval Oz, Assistant Secretary for Research & Technology, U.S. Department of Transportation

Day 2 Keynote and Fireside Chat

MODERATOR **Christos Xenophontos, Assistant Director, Administrative Services, RIDOT**

SESSION DESCRIPTION

This keynote and fireside chat featured **Seval Oz**, the U.S. Department of Transportation's Assistant Secretary for Research and Technology, with a background spanning autonomous mobility, Google X's self-driving car program, intelligent transportation systems, AI, Advanced Driver Assistance Systems (ADAS), and digital transformation. The session focused on how AI can move beyond isolated demonstrations and become part of the operating foundation of the national transportation system.

The keynote was followed by a fireside conversation covering her vision for mobility in 2040, transportation workforce needs, major USDOT initiatives, the Genesis Mission, autonomous vehicle deployment, interstate data interoperability, and the opportunities presented by digital twins.

SESSION HIGHLIGHTS & KEY TAKEAWAYS

- **Mobility 2040 is not primarily about autonomous vehicles or AI.** It is about building a transportation system capable of sensing what is happening, understanding what it means, communicating across boundaries, anticipating what happens next, and acting early enough to improve safety, access, reliability, and economic performance.
- **AI can help treat transportation as an integrated system rather than a collection of independent assets and modes.** Growing computational capacity creates the possibility of treating the transportation network as a much more integrated optimization problem rather than separately managing roads, transit, freight, ports, aviation, and other systems.
- **Mobility 2040 should ultimately be measured by human outcomes.** Oz framed the future not merely around technology, but around whether people—particularly those in rural and underserved areas—can reach jobs, hospitals, schools, family, and essential services.
- **The goal should increasingly be prevention rather than mitigation.** Oz emphasized a shift from reducing the severity of crashes to using AI, sensor fusion, prediction, and communication systems to prevent collisions altogether.
- **Interoperability is foundational.** States cannot operate as isolated digital islands if vehicles, freight, travelers, and incidents routinely cross state boundaries. Oz repeatedly emphasized interoperable transportation operating centers, shared data feeds, common APIs, and corridor-level coordination.
- **Do not wait for perfect national standards before acting.** States can develop "good enough" common language and interfaces now, demonstrate interoperability, and allow more formal national standards to continually evolve.
- **The federal role is to orchestrate.** Oz described USDOT as a funder, testbed provider, research coordinator, and convenor of states, universities, industry, and other federal agencies.
- **Research needs to move out of isolated "sandboxes."** Oz repeatedly called for research, university work, state pilots, industry data, and federal initiatives to be connected so successful concepts can move into national deployment.
- **Autonomous transportation will depend as much on digital infrastructure and regulation as on vehicle technology.** Oz argued that autonomous vehicles do not necessarily require wholesale rebuilding of physical infrastructure, but they will benefit enormously from interoperable digital infrastructure, reliable data, communication systems, and consistent operating environments.
- **Transportation workforce development is a core part of the technology strategy.** Oz's vision for 2040 requires transportation professionals who are as comfortable with data science and computational thinking as they are with traditional infrastructure disciplines.
- **Universities should integrate rather than silo disciplines.** Civil engineering, structural engineering, construction, data science, software engineering, decision science, materials science, and transportation operations will need much more interdisciplinary education and research.
- **Real-world testbeds matter.** Students, researchers, states, and companies need access to actual transportation systems and equipment so they can learn by solving practical problems rather than only conducting isolated laboratory research.

Seval Oz – Assistant Secretary for Research & Technology, U.S. Department of Transportation

Keynote Presentation

BRIDGING THE DIGITAL AND PHYSICAL WORLDS

Oz noted that computer vision, combined with machine learning and analytics are contributing to improvements in transportation safety and mobility, and enabling acceleration in development of autonomous vehicles at scale. However, Oz argued that while AI already performs impressively in digital environments it is not yet able to account for and respond to real world conditions such as temporary detours; unexpected obstacles; changing road conditions; human behavior; and weather. A model that performs well on standard routing or prediction may fail when circumstances depart from its training environment. The next generation of transportation AI must combine digital knowledge with real-world context. She predicted that generative AI will increasingly evolve toward what she described as inference AI, capable of becoming more of a real-time execution engine.



Seval Oz, Assistant Secretary for Research and Technology, USDOT — Day 2 keynote speaker.

TRANSPORTATION APPLICATIONS OF AI

Oz outlined a wide range of applications that she believes will increasingly affect transportation:

- **Adaptive Traffic Management** – enabling dynamic signal timing, prioritization of emergency vehicles; and use of real time forecasts to recommend or implement traveler routing
- **Autonomous Freight Movement** – self-driving trucks using high-definition radar and computer vision for middle-mile and industrial freight applications.
- **Last Mile Delivery** – aerial drones and small ground robots using edge AI to navigate unexpected obstacles and provide last-mile delivery in urban and suburban environments.
- **Infrastructure Inspection** – use of AI-supported scanning for road, bridge, and rail inspections to detect cracking, fatigue, structural defects, and debris.
- **Driver Assistance** – advanced technologies that monitor driver fatigue and distraction, vehicle surroundings, and pedestrian trajectories. Sensor fusion can process these conditions faster than human reaction time and potentially trigger emergency braking before a driver would react.
- **Predictive Operations** – improved prediction of weather effects, congestion, lane closures, emergency conditions, freight disruptions, and bottlenecks to allow agencies to shift from reactive response to preemptive system management.
- **Safety Performance** – use of increasingly sophisticated sensing, prediction, and automated intervention to prevent collisions across all transportation modes.

USDOT'S ROLE IN ACCELERATING AI DEPLOYMENT

Oz described USDOT's role as serving as a primary funder, testbed provider, research coordinator, data and interoperability convener, and deployment catalyst. She emphasized the importance of collaboration with state DOTs, universities, industry, other federal agencies, and national research institutions. The aim is to connect federal funding and coordination, universities and other testbed providers, and practitioners and industry capable of deployment. This creates a

continuous cycle in which data, research, testing, implementation, and learning feed one another rather than operating independently.

NEED FOR INTEROPERABILITY

A major focus of both the keynote and fireside conversation was an emerging corridor-based approach to transportation data interoperability. She noted that air traffic systems routinely have a handoff from one regional control system to another whereas road transportation generally lacks an equivalent digital handoff. Her vision is that transportation operating centers in neighboring states should ultimately be able to communicate seamlessly so vehicles and travelers can receive consistent information regardless of jurisdiction

Oz stressed that states do not need to wait to move forward with building interoperability prior to establishment of formal standards. They can begin with creating compatible Application Programming Interfaces (APIs), identifying common data elements, defining common terminology, and agreeing on latency requirements.

Oz tied transportation interoperability to resilience and national security. She argued that a connected transportation system should not depend on a single communication technology or point of failure. This broader digital backbone could support freight, emergency operations, autonomous mobility, and national supply-chain resilience.

THE GENESIS MISSION

Oz also discussed USDOT's participation in the Genesis Mission, which she presented as a Department of Energy (DOE)-led, major cross-agency effort intended to move research and science more directly into national-scale application. This goal is to build an integrated AI platform to harness Federal scientific datasets, train scientific foundation models and create AI agents to test new hypotheses, automate research workflows, and accelerate scientific breakthroughs.

USDOT became part of an initial group of agencies involved in the effort, bringing transportation-related projects such as complementary Positioning, Navigation and Timing (PNT) and AI-enabled prediction and analysis. Oz described transportation data from state operating centers ultimately contributing to broader secure and anonymized data platforms that AI can search, filter, index, and analyze. Potential applications include anticipating weather disruptions; lane closures; collision risks; timing problems; freight disruptions; and conditions affecting autonomous vehicle deployment. This initiative provides the opportunity to use AI to transform how we build, operate, maintain transportation systems, and to integrate interoperability into the backbone grid.

LOOKING TOWARD 2030

When discussing the nearer-term future, Oz identified several areas where she expects substantial progress by around 2030:

- interoperable V2X networks;
- Level 4 autonomous middle-mile freight;
- a federal framework for autonomous vehicles;
- beyond-visual-line-of-sight (BVLOS) commercial drone delivery;
- AI-native transportation management;
- digital twins;
- expanded in-cabin monitoring; and

- standardized autonomous emergency braking systems

WORKFORCE DEVELOPMENT

Oz said her 2040 workforce vision is for transportation professionals to be **as** comfortable with data, data science, and physical science as they are with steel and concrete. Achieving this vision will require more than importing technology talent into transportation agencies. New educational models, internal staff development and partnering with industry will be required.

Oz advocated expanding the skills of existing and future transportation professionals through integrating curricula on decision science, computational analysis, and systems thinking into traditional civil and structural engineering education. She also supported giving universities access to actual infrastructure, vehicles, buses, corridors, and operating environments to provide testbeds that allow students to learn by solving real transportation problems in real time. She pointed to the value of industry providing equipment and public agencies providing test environments so research is directly connected with deployment.

Oz mentioned several existing mechanisms that can contribute to workforce development, including:

- the ITS Joint Program Office Professional Capacity Building program;
- SMART grants;
- the University Transportation Center program; and
- partnerships among universities, state agencies, federal programs, and industry.

She also argued that these programs should share data and coordinate more extensively rather than allowing similar research efforts to remain isolated from one another.

Fireside Chat and Q&A

VISION FOR MOBILITY 2040

Xenophontos asked Oz to describe what mobility could look like in 2040.

Her 2040 vision included a transportation system that “talks to itself”; detects anomalous patterns; recognizes problems; automates responses; coordinates across jurisdictions; supports widespread autonomous vehicles; moves freight faster and more efficiently; reduces transportation costs; improves access; and gives people more control over how they use travel time.

She emphasized that none of that succeeds without the workforce necessary to design, operate, maintain, evaluate, and improve the systems.

WHAT CURRENT INITIATIVE MATTERS MOST TO MOBILITY 2040?

When asked which current initiative could have the greatest impact on Mobility 2040, Oz highlighted the **corridors initiative**, particularly its ability to prepare the digital foundation for large-scale autonomous mobility. The focus is on getting neighboring states to harmonize and exchange real-time data so their transportation operating centers can communicate across interstate corridors. This initiative embeds digitalization preparedness for launching AVs. FHWA is providing $200K per state, and OST is providing an additional $10 million in funding.

WHAT ARE THE BIGGEST CONSTRAINTS TO PROGRESS?

Major constraints include fragmented regulations; inconsistent data; state-by-state systems; funding for digital infrastructure; workforce capacity; the pace of technology evolution; and coordination between government and industry.

Technology evolves exponentially while organizations—and particularly government institutions—often evolve much more slowly.

HOW LONG WILL IT TAKE FOR AVS TO SCALE?

Drawing on her Google self-driving experience, Oz said that industry technology advanced much faster than the regulatory environment converged. The result has been a patchwork of state regulations that complicates national deployment.

Oz acknowledged that one of the hardest stages will be the coexistence of human-driven vehicles; and computer-driven vehicles. Humans will need to learn how automated vehicles behave, while autonomous systems will need to operate safely amid unpredictable human behavior. She suggested that this mixed environment will likely persist for the next five to ten years before automated driving becomes much more commonplace and socially familiar. Her broader message was that agencies cannot wait for that future to arrive before beginning to learn. They need to test, collect data, develop rules, and build infrastructure now.

DIGITAL TWINS AND RHODE ISLAND

Near the end of the session, Xenophontos discussed a possible future RIDOT symposium focused on digital twins and floated the idea of Rhode Island creating a digital twin of its transportation network.

Because Rhode Island is relatively small, the state could potentially develop an integrated representation of highways; utilities; roadway work; toll systems; and other transportation infrastructure. That statewide twin could eventually communicate with equivalent systems in Massachusetts and Connecticut.

Oz strongly endorsed the idea, particularly the possibility of demonstrating a digital twin border-to-border and showing how a model could propagate across jurisdictions.

DAY 2 · SESSION 02

AI in Planning and Project Development

Technical Session 3

MODERATOR **Craig Thor, Chief Scientist, FHWA Turner-Fairbank Highway Research Center**

SESSION DESCRIPTION

This technical session examined how AI, computer vision, machine learning, geospatial analytics, and emerging generative and agentic tools can improve transportation planning, asset management, project development, and decision support. The session featured:

- **Greg Ciparelli, Chief Data Officer at Connecticut DOT,** describing CTDOT's PRISM pavement-marking inventory and management system, which combines field data, GIS, imagery, LiDAR, machine learning, and forecasting;

- **Sophie Abo of the USDOT Volpe Center**, describing the AI for Transportation Planning and Design (AITPD) initiative, which supports small businesses developing AI-enabled multimodal data and planning tools; and
- **Grace Sammartino of Esri,** describing how geospatial AI, computer vision, deep learning, generative AI, and emerging agentic capabilities are being incorporated into enterprise GIS workflows.

SESSION TAKEAWAYS

- **AI can turn labor-intensive inventories into scalable digital assets.** CTDOT used computer vision and machine learning to extract and classify pavement markings that otherwise could have taken months or years to digitize manually.
- **Computer vision is becoming more accessible and reusable.** Esri described dozens of pre-trained models and increasingly end-to-end workflows that allow agencies to apply AI to imagery and LiDAR without developing every model from scratch.
- **Unstructured documents are another important AI opportunity.** Information locked in plans, as-builts, design documents, and other files can increasingly be extracted and geolocated.
- **GIS can provide the organizing backbone.** Across all three presentations, location was the common denominator that allowed imagery, LiDAR, roadway attributes, asset inventories, maintenance plans, project schedules, demographic data, and other information to be combined.
- **Good AI depends on good source data, but no source is perfect.** Imagery age, snow and ice, degraded pavement markings, poor rural imagery, inconsistent databases, and imperfect training samples all affect model performance.
- **Multiple independent data sources can strengthen validation.** Aerial imagery, street-level imagery, LiDAR, field observations, and other sources can be used to cross-check model outputs.
- **Validation should be risk-based rather than perfection-based.** The panel cautioned against rejecting an application merely because it is not 100% accurate. A 70% model may be unacceptable for regulatory speed-limit enforcement but still valuable for screening or prioritization.
- **Guardrails around data are critical for generative and agentic applications.** The Caltrans example intentionally returned no result when asked for information that did not exist in the authorized dataset, demonstrating the value of limiting AI to trusted information.
- **AI tools need to fit existing agency workflows.** USDOT's AITPD experience showed that agencies generally do not want raw data or sophisticated standalone tools; they want applications that fit naturally into planning, programming, and design processes.
- **AI should address real practitioner needs rather than technology for technology's sake.** Direct engagement with local agencies has been essential in the AITPD initiative.
- **The real value comes from integrating AI outputs into existing business processes.** A map or model alone is not enough. CTDOT connected the pavement-marking inventory to cost estimating, maintenance programming, pavement projects, capital projects, and future degradation forecasting.
- **The long-term opportunity is democratization of analysis.** Panelists repeatedly returned to making sophisticated information and analytic capabilities available to more employees, municipalities, regional agencies, and other partners rather than concentrating expertise in a small number of specialists.
- **Breaking down organizational and data silos may be more transformative than any individual AI model.** This was one of the panel's clearest 2040 themes.

Greg Ciparelli — Transportation Planning Director and Chief Data Officer, Connecticut Department of Transportation

Pavement Marking Management

Ciparelli described development of PRISM — CTDOT's Pavement Marking Retroreflectivity Inventory Scheduling and Management System. PRISM was developed in partnership with the University of Connecticut. The system was based on computer vision and machine learning methods. It created a full statewide inventory of pavement markings, including a condition index based on retro reflectivity measures. The system can be used to estimate funding needs for pavement marking maintenance and identify potential pavement marking projects.

BUILDING THE INVENTORY

To build the pavement marking inventory, CTDOT equipped a vehicle with a retroreflectivity measurement device and had UConn drive the state highway system in both directions. That collection provided GPS-located pavement-marking positions; and measured retroreflectivity values.

The next step was to extract pavement marking features that could be placed on the agency's linear referencing system (LRS) and classify the extracted features by type of marking (e.g., solid white, dashed white, double yellow, etc.). This step was accomplished using a computer vision machine learning model applied to statewide high-resolution aerial imagery and LiDAR collected in 2023.

The process included multiple steps: segmenting imagery; extracting potential pavement-marking features; using LiDAR intensity and other attributes to improve identification; classifying the extracted features; smoothing and correcting the geometry; and assigning final pavement-marking types and attributes.

The network was divided into 0.1-mile segments and buffered to create roadway envelopes. Pavement-marking features and retroreflectivity values could then be assigned to those roadway segments. features also contained information such as width, length, and spatial position.

Ciparelli emphasized that the model required iterative human feedback. Researchers adjusted variables and classifications as they encountered errors.



CTDOT's pavement-marking feature extraction workflow, presented by Greg Ciparelli.

The combined methodology achieved approximately 97–98% accuracy for feature extraction and classification. Errors still occurred, requiring manual cleanup for misclassifications. The model does not eliminate Quality Assurance (QA). It dramatically reduces the amount of work humans need to perform by narrowing a huge statewide digitization problem into a much smaller set of exceptions and corrections.

Ciparelli noted that the extraction model was developed using 2023 imagery, while transportation assets continue to change. CTDOT therefore needs to determine how the model will perform against later imagery collections and how frequently extraction should be repeated.

From a planning perspective, that transforms an inventory into something that can support investment decisions.

USING PRISM FOR PROJECT PLANNING AND COST ESTIMATING

Maintenance staff can select a proposed pavement-marking project and automatically calculate costs associated with pavement marking installation, accounting for marking type, traffic control, and other project costs. This replaces ad hoc manual calculations with consistent project-cost information.

Because the pavement markings are located on the LRS, staff can check proposed marking work against other programmed investments. For example, if the pavement is scheduled for reconstruction next year, it may not make sense to install a long-life pavement marking today. Maintenance and capital programs can therefore be coordinated rather than independently replacing the same asset multiple times.

PREDICTING PAVEMENT-MARKING DEGRADATION

CTDOT's next step is forecasting. The agency is developing a degradation model using 26 variables, including factors such as precipitation; snow removal; traffic volumes; and truck volumes.

The intent is to predict when markings are likely to deteriorate below desired levels and therefore improve three-, four-, and five-year programming.

Sophie Abo — Transportation Planner, USDOT Volpe Center

THE AI FOR TRANSPORTATION PLANNING AND DESIGN INITIATIVE

Abo presented the **AI for Transportation Planning and Design (AITPD) initiative**, a multi-year USDOT effort managed by Volpe on behalf of the ITS JPO. Funding for AITPD is provided through the federal **Small Business Innovation Research (SBIR)** program, which provides seed funding for companies developing technologies in markets that remain underdeveloped.

The initiative began in response to a common local-agency challenge: collecting and maintaining detailed multimodal transportation data can be extremely expensive and time-consuming.

Abo cited sidewalk inventory work as an example, noting that manual mapping can cost roughly $100–$800 per mile, while the resulting information may become outdated relatively quickly.

AITPD seeks to develop AI-enabled planning software based on scalable methods for collecting accurate, high-resolution data. The goal is to improve systems planning; project design; early project development; and agency decision support.

There have been two phases of the initiative.

In Phase One, 12 small businesses received $200,000 awards for efforts focusing on AI-enabled multimodal data collection, extraction, and fusion.

The companies used very different approaches, including:

- sensors and edge computing mounted on bicycles to identify near misses;
- computer vision capable of identifying more than 100 right-of-way asset types;
- highly detailed 3D curb modeling;
- and imagery/LiDAR approaches for pavement markings and retroreflectivity.

The purpose was not to mandate one technical solution, but to see which approaches could successfully generate transportation information that agencies often struggle to collect.

Five companies advanced to Phase Two (currently in progress). Each received a two-year, $1.7 million award to:

- scale data-collection methods;
- expand to additional geographies;
- continue improving data fusion;
- develop software products; and
- translate raw information into useful agency decision support.

Several tools are also incorporating LLM-based interfaces so users can query data conversationally.

EARLY LESSONS FROM AITPD

While Phase Two of AITPD is ongoing, Abo shared some early lessons from the initiative:

- **AI Can Scale Multimodal Data Collection.** The program has demonstrated that AI can improve transportation data generation, collection, and fusion at scales ranging from individual cities to regional or potentially national applications.
- **Aerial Imagery and Street-Level Imagery Have Different Strengths.** Aerial imagery provides broad geographic coverage whereas street-level imagery can provide much greater detail for sidewalks, curbs, roadway assets, behavior, and other fine-grained transportation information. The appropriate source therefore depends on the use case.
- **Results are Impacted by Weather.** Snow and ice created major problems during Phase One. Weather affected both the ability to collect usable data; and the ability of AI models to identify intended asset and behavior types. This is a practical reminder that field-based AI applications must be evaluated under the actual environmental conditions in which transportation agencies operate.
- **Agencies Want Workflow Integration.** Agencies want tools that connect directly with existing processes for planning, programming, and project design. They are not interested in raw data dumps or sophisticated but isolated software.
- **Practitioner Partnerships Are Essential.** Direct participation by municipalities and other transportation practitioners helps ensure that the technology addresses genuine needs.

Ultimately, USDOT hopes AITPD will produce detailed and continuously improving multimodal datasets and tools that can help agencies to identify safety gaps; identify critical needs; prioritize projects; and improve planning and design decisions.

Grace Sammartino — Senior Account Executive, Esri

AI in Planning and Project Development

AI WITHIN ARCGIS

Sammartino provided a private-industry perspective on how AI is increasingly becoming embedded throughout the **ArcGIS geospatial platform**. Her presentation emphasized that GIS should not be viewed simply as a mapping application. She described it instead as a system for managing, integrating, analyzing, and presenting many types of transportation information. Her focus was on how AI complements existing geospatial workflows rather than treating it as a separate product.

Sammartino explained that machine-learning capabilities have existed in ArcGIS for years, but historically they were mainly accessible to specialized data scientists. More recent developments are intended to make these tools usable by a much broader audience through preconfigured environments; pre-trained models; conversational interfaces; and low-code/no-code applications.

Sammartino described AI as increasingly being distributed throughout the ArcGIS suite rather than residing in one specialized application. She described three broad layers of the platform architecture: data and tools; services and analytics; and presentation (including maps and applications). AI techniques can operate across these layers rather than as an isolated system.

There are currently over 70 pre-trained models that can be applied to imagery and other geospatial data. Examples include imagery analysis, LiDAR and 3D extraction, and object detection.

Sammartino also highlighted the opportunity to extract useful geographic information from unstructured documents such as as-built plans and make it useable in a GIS environment.

She described an emerging tool called Location Embeddings, intended to accelerate analysis that combines geographic; socioeconomic; and demographic information. This could support more complex planning analyses without requiring analysts to manually assemble and transform every dataset.

GEOSPATIAL AI APPLICATIONS

Sammartino rapidly walked through numerous examples of geospatial AI applications including predictive crash analysis, survey development, automated 3D digital twins, asset extraction from street-level imagery, pavement distress detection, and analysis of post-event imagery to inform emergency response activities.

Sammartino predicted that geo-prompting—asking natural-language questions of imagery and spatial information—will become an increasingly important skill for transportation staff. For example, staff might ask a system to inspect aerial imagery for damaged structures and debris and receive a conversational response based on what the imagery contained.

This suggests a shift in workforce capability: Employees may not need to build every model themselves, but they will need to know what to ask, what data to use, how to interpret results, and when additional verification is needed.

Sammartino stressed that AI readiness also depends on metadata. Esri is developing an assistant intended to help describe and structure datasets so they are easier for people and AI systems to understand.

EMERGING CAPABILITIES

Sammartino closed by describing work to make ArcGIS interoperable with emerging agentic AI platforms. Her description suggested an environment where users could call ArcGIS capabilities from other AI systems; bring outside AI agents into a spatial environment; develop agentic mapping applications; and use natural language to access geospatial tools and data. She emphasized that these emerging applications need to operate within controlled and trusted data environments.

Discussion and Q&A

HOW CAN AGENCIES VALIDATE AI-GENERATED DATA?

Esri: A technical Esri representative emphasized that AI accuracy depends heavily on the training samples. If training examples are poor, results will be poor—essentially the familiar "garbage in, garbage out" principle .He described running AI across 10,000 images to identify those containing traffic signs and noted that the system occasionally identified a sign a human initially overlooked. The broader point was that computer vision can sometimes outperform humans in repetitive image review, but performance remains tied to training quality. Sammartino added that human-in-the-loop review remains necessary.

Ciparelli/CTDOT: Ciparelli suggested that staff understand limitations of model output and make risk-based decisions about acceptable accuracy thresholds and the level of required validation based on the use case. A lower-confidence model may be acceptable for screening but unacceptable for regulatory or safety-critical action. Ciparelli also pointed out that traditional datasets such as HPMS have their own simplifications and limitations, so AI-generated information should be evaluated relative to the quality of existing alternatives—not against an imaginary perfect dataset.

Abo/Volpe: Abo added that using multiple data type**s** can strengthen validation. Examples include reserving one data source—such as LiDAR, aerial imagery, or street-level imagery—to validate outputs derived from another.

WHAT IS THE BIGGEST OPPORTUNITY LOOKING TOWARD 2040?

Thor closed by asking the panel what excites them most as these technologies mature over the next 5, 10, or 15 years.

Sammartino/Esri: The ability of AI to help break down organizational and data silos.

Ciparelli/CTDOT: Democratization of information and analysis - CTDOT has thousands of employees and strong knowledge of its own state network, but less visibility into local roads, parks, forests, and facilities controlled by other agencies. As analytic tools become more accessible, municipalities and other organizations may be able to approach the DOT with better-informed ideas.

Abo/Volpe: Ivo was reluctant to predict 2040 given how quickly AI is evolving, but her nearer-term aspiration was for the advances in computer vision to become more widely available to agencies of all sizes—not only highly capable state DOTs.

DAY 2 · SESSION 03

AI in Asset Management, Predictive Maintenance, and Inspection

Technical Session 4

MODERATOR **Christopher Baxter, Professor, University of Rhode Island**

SESSION DESCRIPTION

This technical session focused on how AI, digital twins, structural health monitoring, sensors, and asset-management systems can help transportation agencies move from periodic inspection and reactive maintenance toward continuous monitoring, earlier detection of deterioration, and more proactive, risk-informed intervention..

The speakers explored different parts of a potential solution:

- **Donovan Guin of IBM Consulting** framed the need for continuously monitored civil infrastructure and discussed RIDOT's bridge challenges and the concept of an AI-enabled digital twin.
- **Scott Hensley of IBM Consulting** explained how sensor-derived alerts can be connected to IBM Maximo so that anomaly detection leads to documented inspections, work orders, corrective actions, and organizational learning.
- **Arnulf Hagen, CEO of Aidalos** (Norway) demonstrated an engineering-focused AI system capable of reviewing sensor data, inspection reports, photographs, and outside contextual information; identifying anomalous structural behavior; and supporting engineering investigation.
- **Mehrshad Amini, Assistant Professor at the University of Rhode Island**, described URI's new multi-hazard structural monitoring and digital-twin initiative and its Warren Bridge testbed, combining sensors, virtual models, physics-based analysis, AI, and workforce development.

A major theme throughout the session was that **AI should not replace engineers or inspectors**. Instead, its role is to provide continuous attention, identify patterns humans may not have time to detect, cross-check inspection information, organize evidence, and direct scarce engineering expertise toward the situations that most deserve attention.

SESSION TAKEAWAYS

- **Bridge inspection is fundamentally constrained by staff capacity and inspection frequency.** Transportation agencies cannot have engineers continuously watching every structure. AI-enabled monitoring offers a way to keep a "finger on the pulse" of infrastructure between inspections.
- **Inspection reports and photographs can be analyzed together.** AI can compare written reports against photographic evidence and potentially identify conditions that were visible but not documented.
- **Sensors alone are not enough.** Raw sensor dashboards may identify movement or changes, but someone still has to determine whether the change is meaningful and what should happen next.
- **The objective is to shift from reactive to proactive asset management.** Rather than discovering serious deterioration through periodic inspections, agencies could continuously screen for unusual behavior and intervene earlier.

- **Avoid alert fatigue.** Continuous monitoring will only help agencies if the system can limit false positives and make each flagged event easy enough to investigate that staff do not begin ignoring alerts.
- **Engineering AI needs to express uncertainty.** AI should distinguish among conclusions that are possible, plausible, likely, or unlikely and leave the final decision to a qualified human.
- **AI can serve as an additional quality-assurance layer.** The presenters emphasized that this is not intended to replace inspection; it can flag inconsistencies, missing observations, or trends requiring further human review.
- **Data-driven and physics-based digital twins serve different purposes.** A data-driven twin can monitor and visualize changing conditions, while a physics-based twin adds finite-element or other physical models that can support response prediction, damage prediction, and "what-if" analyses.
- **Universities can provide both innovation and workforce development.** URI is using the Warren Bridge as a real-world testbed while involving undergraduate and graduate students in sensor installation, modeling, coding, and research.
- **The business outcome matters more than the technology.** The closing remarks returned to a symposium-wide theme: AI should be judged by whether it improves safety, reliability, infrastructure stewardship, and outcomes for people—not by the novelty of the technology.

Donovan Guin — US Public Transportation Lead, IBM

Digital Twins for Civil Infrastructure

Guin began by describing the basic challenge of maintaining aging civil infrastructure when agencies cannot continuously inspect every critical structure.

His argument was that visual inspection has inherent limitations because serious structural deterioration may be masked by debris, occur in locations that are not visible, or may progress between formal inspections. He suggested that if movement or other structural response could be continuously monitored, abnormal behavior might be detected earlier than through periodic inspection alone.

Guin characterized conventional inspection as something of a **"**whack-a-mole" problem: engineers visit structures periodically and hope to identify the right problem at the right time.

The proposed alternative is an AI-supported system that can continually monitor structures; identify where attention may be needed; help determine what type of attention is appropriate; and help agencies intervene at the most cost-effective point in the asset lifecycle. This approach could allow agencies to stretch limited funding without compromising safety.

Guin emphasized that bridge deterioration can be affected by various factors including traffic composition and speed, construction activity, and environmental conditions, and suggested that future versions of this concept could connect bridge monitoring with transportation-system operations data such as ATMS information. The larger vision is therefore not simply a sensorized bridge but an integrated digital environment that understands what is happening to the structure and what external conditions may be causing it.

LESSONS FROM INTERNATIONAL WORK

Guin referenced IBM work with infrastructure organizations in Italy, Denmark, and the Port of Rotterdam. In Italy, BIM and asset-management information are being combined with drone-

based visual inspections, where visual AI can identify deterioration and trigger follow-up work. The Port of Rotterdam example helped inspire a different model: a real-time digital twin that combines infrastructure, operational, and environmental data.

Guin introduced the concept for a RIDOT pilot (presented later in the session) that builds on this experience, combining digital infrastructure representation, sensor time series data, traffic/environmental context information, AI, and asset-management workflow.

Scott Hensley — Associated Partner, IBM

Hensley's presentation focused on actions taken by the agency once anomalies are detected. He stressed that collecting sensor data and creating a digital twin are useful, but neither alone completes the maintenance process. An alert must be received, assigned, investigated, documented, resolved, and used to improve future decisions.

An AI-generated anomaly can be associated with a specific bridge component and used to initiate an appropriate response such as an additional inspection, a maintenance work order, replacement of a faulty sensor, or a repair or other corrective activity.

Creating a closed learning loop takes this one step further, building on the experience to improve future recommendations. If a similar condition occurs later, the system could use the previous response as part of its guidance.

Hensley noted that whereas a simple condition-monitoring platform tells the agency something happened, an integrated asset-management environment tells the agency where it happened, who needs to respond, what was previously done in similar circumstances, what actions are available, and whether the issue was ultimately closed. The objective is to convert an AI insight into something that can actually be executed by people responsible for the bridge.

Arnulf Hagen, Founder and CEO, Aidalos

AI Agents in Bridge Asset Management

Hagen demonstrated an engineering-focused AI system intended to continuously analyze infrastructure information.

He provided background on Aidalos and characterized the company's technology as a neurosymbolic, grounded cognitive architecture built by engineers for engineers.

The AI system is intended to act as a persistent analytical partner rather than merely as a chatbot.

NORWEGIAN BRIDGE EXAMPLE

Hagen described an earlier Norwegian bridge case that helped inspire the system. Sensor data showed structural behavior changing over several days. Initially, inspectors found nothing visually obvious. The engineering team remained concerned and pushed for another inspection. The second inspection revealed a serious failure at one of four support points, and the bridge was closed immediately.

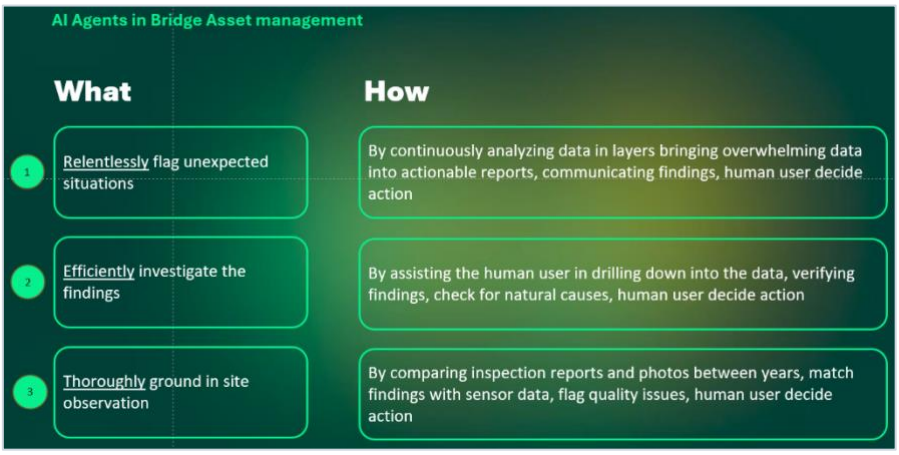
When the data were later analyzed using the newer AI capabilities, the system reportedly could identify the developing abnormal behavior weeks earlier. The important signal was not simply an extreme measurement - it was the progressive change in the pattern. Hagen stressed that engineers had possessed the data but lacked the capacity to continuously search it for those subtle patterns.

A GROUNDED MULTI-AGENT ENGINEERING SYSTEM

The Aidalos system monitors a range of inputs including sensor time series data, inspection reports, photographs, structural calculations, weather data, seismic information, and construction information. Rather than having one generic model provide an answer, the architecture uses specialized agents and engineering tools.

A human remains in dialogue with the system. They can pose a number of questions to the system – for example:

- Did something unusual happen?
- Is this anomaly significant?
- Is additional investigation warranted?
- How much effort should be spent?
- What evidence supports each explanation?
- What engineering response should be considered?



Multi-agent engineering analysis of a bridge in the Aidalos system, presented by Arnulf Hagen.

RIDOT SENSOR EXAMPLES

Hagan presented a RIDOT pilot application using the Aidalos system. Using real bridge sensor information, Hagen demonstrated a workflow where the system identified unusual behavior concentrated around particular spans. One anomaly occurred during the daytime. Rather than immediately concluding the bridge had deteriorated, the system investigated outside information and identified demolition work on the adjacent structure as a plausible explanation.

Arnold stressed an important distinction: the system did not claim that explanation was definitely true. Rather, It identified it as the most compelling explanation, leaving the engineer to make the decision.

That example illustrated what Arnold called the need for an AI that can doubt. In engineering, a system that confidently states uncertain conclusions could be dangerous.

For a second bridge, the team provided the AI with several years of inspection reports and photographs. The system reviewed the materials and characterized the situation as serious and recommended actions including an updated load rating and further structural evaluation.

When Hagen called to report that the AI indicated the bridge should be closed, RIDOT contacted its bridge engineering staff. The response was that RIDOT had already closed the bridge the previous week. In other words, the AI independently reached a conclusion consistent with the agency's actual engineering action. This served as a validation test rather than a case in which AI itself caused the closure.

AVOIDING FALSE POSITIVES

Arnold emphasized that continuous monitoring could become counterproductive if engineers are overwhelmed by alarms.

The system therefore needs to flag unexpected behavior; make investigation efficient; suppress or explain benign conditions; qualify uncertainty; and escalate only conditions deserving human attention.

Mehrshad Amini — Assistant Professor, University of Rhode Island

Multi-hazard Structural Health Monitoring, Bridging Sensing, Data Analytics, and Digital Twin Systems

Amini presented URI's newly established multi-hazard structural monitoring and digital-twin initiative. The initiative combines structural engineering, sensing, data management, AI, virtual modeling, physics-based analysis, transportation agency collaboration, and student/workforce development.

Amini noted the broader trend toward more U.S. bridges being categorized in fair rather than good condition. Rhode Island also presents specific environmental challenges because of its long coastline and exposure to sea-level rise, flooding, scour, erosion, and other coastal hazards. He therefore framed the digital twin not solely as a structural-condition system but as a multi-hazard decision-support framework.

WHAT IS A DIGITAL TWIN?

Amini emphasized that a digital twin is more than a 3D visualization.

He described it as a virtual model that:

- represents a physical asset;
- is dynamically updated using information from the physical asset;
- can support prediction and decision-making; and
- maintains interaction between the physical and virtual worlds.

He also emphasized three requirements surrounding the model: verification, validation; and uncertainty quantification. At agency scale, security and ethics are additional considerations.

Amini differentiated between data-driven digital twins that provide a virtual representation of the facility and physics-based digital twins that incorporate finite-element or other structural models. A data-driven digital twin may be appropriate for a monitoring dashboard, but a physics-based digital twin would be needed to provide "what if" loading scenario analysis capabilities.

URI CENTER OBJECTIVES

Amini described five broad objectives:

- Establish an advanced structural-health-monitoring and digital-twin facility at URI.
- Develop a real-world instrumented bridge testbed.
- Create data-driven and physics-based digital-twin models.
- Train undergraduate and graduate students.
- Sustain partnerships and workforce-development impacts over time.

WARREN BRIDGE TESTBED

The primary real-world testbed is the Warren Bridge on Route 114, connecting Warren and Barrington. Amini noted that the bridge is part of an evacuation route, increasing its importance during hurricanes and other emergencies.

The research team is deploying or planning multiple sensor types including Global Navigation Satellite System (GNSS) sensors, accelerometers, strain gauges, sonar depth sensors for scour, water-level sensors, and cameras capable of supporting computer-vision analysis of traffic. The GNSS sensors support long-term displacement monitoring, while accelerometers will capture higher-frequency dynamic bridge response. The sensors can feed data through APIs into the research environment.

VIRTUAL AND PHYSICS-BASED MODELS

The URI team is developing a virtual bridge representation using 3D parametric bridge information modeling (BIM) software and incorporating sensor locations into that environment.

The next stage is to develop a physics-based finite-element model capable of supporting damage prediction, response prediction, structural characterization, and “what-if” scenarios.

The team is validating its physics-based methods using published monitoring data from the Z24 bridge in Switzerland before applying them to the Warren Bridge.

WORKFORCE DEVELOPMENT

Amini emphasized workforce development throughout the presentation.

Students are involved directly in sensor selection, installation, data management, coding, modeling, and field implementation. He highlighted one student who began as an undergraduate researcher, moved into a RIDOT internship associated with the project, and planned to begin graduate study. This illustrates how transportation agencies and universities can build a pipeline of professionals fluent in both civil engineering and digital technologies.

Discussion and Q&A

CAN AI HELP EVALUATE FATIGUE AND ADDED TRAFFIC LOADING?

Baxter asked Hagen whether the bridge monitoring work could ultimately be used to assess fatigue associated with changing traffic and increased loading.

Hagen explained that the architecture is designed so specialized engineering tools can be added as needed. Fatigue analysis is one such capability under development. However, he emphasized that doing this on an existing older bridge is more difficult than designing monitoring into a new bridge from the outset.

Another technical participant explained that fatigue calculations already exist within their engineering workflow, although they had not yet been fully integrated into the AI tool. The eventual goal is for an engineer to be able to ask the system about fatigue damage at specific locations.

HOW DO YOU ENSURE THE INPUT DATA ARE RELIABLE?

An audience member asked how the Aidalos system handles quality when information comes from a variety of sources, gathered at different points in time, with potentially inconsistent reporting practices.

Hagen agreed that input quality assurance is fundamental. He described several layers of quality control:

- algorithmic quality control of incoming data;
- review of inspection reports;
- comparison of written reports with photographs;
- peer-review agents;
- a contrarian agent intended to challenge conclusions; and
- additional checks within the neurosymbolic architecture.

The system might determine, for example, that the written inspection report is reasonable but a photograph appears to show a condition that was not mentioned. That does not mean the AI overrides the inspector. Instead, it creates another quality-assurance checkpoint and asks a person to investigate.

HUMAN + AI RATHER THAN HUMAN VS. AI

The discussion repeatedly returned to the need to keep humans responsible for engineering decisions. The Aidalos system is intentionally designed to give qualified statements such as “possible”, “plausible”, “likely”, “unlikely”, or “worthy of further investigation.” This contrasts with the false certainty that can occur in general-purpose generative AI.

In a safety-critical engineering context, the AI's job is to organize evidence and support professional judgment—not pretend that uncertainty does not exist.

RIDOT’s closing comments similarly described its AI direction as bringing people and AI together, rather than delegating transportation decisions entirely to machines.

DAY 2 · SESSION 04

From Data to Digital Twins: The Future of Asset Management

Technical Session 5

MODERATOR **Robert Sheehan, Director, FHWA AI Deployment Center of Excellence**

SESSION DESCRIPTION

This session explored data fusion techniques for merging disparate and often siloed data sources, including vehicle telematics, roadside sensors, social media signals, and weather data, into unified analytics platforms.

Presenters discussed methods for cleaning, aligning, and integrating these datasets, and demonstrated how fused data can generate comprehensive insights into network performance, travel patterns, and user behavior that no single data source could reveal on its own. The session featured:

- **Charles Major, Manager of GIS Data Engineering at MassDOT Office of Transportation Planning,** describing MassDOT’s progression from experimental computer-vision pilots to making AI-enabled feature extraction a more routine part of GIS data development. He covered crosswalks, sidewalks, facility footprints, CCTVs, traffic signs, rail right-of-way, historical engineering plans, and future pavement-marking inventories.

- **Giri Venkiteela, Research Scientist at NJDOT's Bureau of Research and Innovation,** discussing New Jersey's approach of starting with AI pilots that can demonstrate value to leadership.
- **Grace Sammartino of ESRI,** who complemented her prior presentation with a discussion of moving data across the facility lifecycle and examples of digital twin applications.

SESSION TAKEAWAYS

- **AI depends on good underlying data.** As DOTs expand their use of AI and machine learning, the demand for accurate, current, and rapidly available transportation data grows rather than diminishes.
- **Traditional GIS data-development methods can be too slow for current agency needs.** Processes that once took six months, a year, or longer are increasingly incompatible with rapidly changing transportation systems and expanding demands from agency business units.
- **Computer vision can create inventories that agencies previously lacked the staffing capacity to build.** Computer vision can identify asset types, capture attributes, classify condition, and turn unstructured historical documents into structured GIS information.
- **Historical engineering plans represent a major untapped data resource.** DOTs possess decades of paper plans, scanned maps, PDFs, and CAD files containing valuable asset information that staff do not have time to manually digitize.
- **Validation remains indispensable.** AI can accelerate data creation, but agencies still need a deliberate quality-assurance process.
- **Start with accessible tools and build staff capability.** MassDOT found value in codeless tools that allowed GIS professionals without data-science backgrounds to learn model training, validation, and feature extraction before progressing to more sophisticated internally developed models.
- **AI implementation can provide workforce-development opportunities.** MassDOT's computer-vision work increased staff understanding of data science and reduced dependence on outside research partners over time.
- **AI can become an institutional-memory system, not merely a tool for summarizing individual documents.** New Jersey's concept is to bring inspection reports, manuals, historical information, imagery, and other bridge information together so staff can query knowledge across documents and over time.
- **Combining AI approaches may outperform reliance on a single model.** For bridge defect detection, New Jersey is testing hybrid approaches involving vision-language models and convolutional neural networks; the speaker reported results approaching the ground-truth classifications.
- **The longer-term vision is AI at the point of inspection.** An inspector could photograph a bridge component and receive immediate information informed not only by the image but also by agency manuals, standards, thresholds, and previous inspection information.
- **Agencies do not need to resolve every technical definition or reach the most sophisticated version of a digital twin before beginning.** They can start with a specific problem, build useful digital information, involve the people who will use it, demonstrate value, and progressively increase sophistication.

Charles Major — Manager of GIS Data Engineering, Massachusetts DOT

Continued development of computer vision based feature extraction workflows for multiple GIS assets

Charles explained that his team is embedded in MassDOT planning and manages substantial portions of the agency's GIS data.

Traditional GIS development could involve months of work to create a dataset – some complex datasets could take six months to a year to develop. The rapid growth of AI and machine learning has increased expectations for data availability and accuracy. That led MassDOT to begin experimenting with computer vision for transportation asset extraction. What began as pilots gradually became a broader strategy, with MassDOT now working toward treating computer vision as a normal tool in its GIS-data-development toolbox.

Major emphasized that improving one GIS dataset can be used for multiple purposes – some of which may not be anticipated. As an example, MassDOT's bridge-clearance information was used for planning motor-coach routing associated with Boston-area activities for the 2026 FIFA World Cup. The lesson is that agencies may underestimate the return from investing in foundational datasets because the information is often reused in applications that were not envisioned when the dataset was first developed. Once an accurate, structured transportation dataset exists, other parts of the agency can find new uses for it.

Major described several examples of how MassDOT is using AI/computer vision to build inventories of crosswalks, traffic signs, sidewalks and pedestrian facilities, and building footprints. They have also used machine learning techniques to extract information on CCTVs from design plans (based on symbology) into a GIS asset layer.

Planned future efforts include a pavement marking inventory, and extracting rail right-of-way information (and possibly additional rail features) from engineering plans and potentially expand the process to additional rail features.

Through research-supported pilots and codeless computer vision platforms, MassDOT has been able to build its staff capabilities in use of computer vision and machine learning techniques. Staff have learned how to: define training areas, classify features, adjust training data, retrain models, validate outputs; and improve results through review and iteration. MassDOT is now developing more of these models internally and relying less exclusively on external research organizations or packaged platforms.

Major summarized the effect as an exponential improvement in the team's data-development capacity by combining its traditional GIS expertise with growing knowledge of data science.

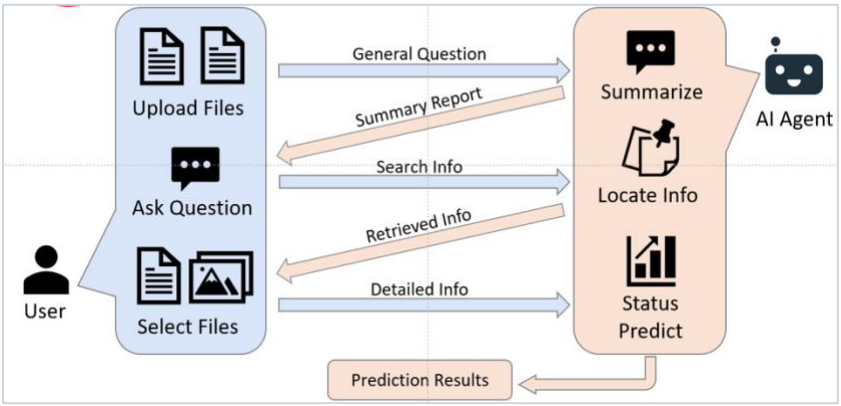
Giri Venkiteela, Innovation Officer, NJ DOT

AI in Asset Management

Venkiteela's presentation covered two AI pilots related to bridges.

AI-ENABLED BRIDGE INSPECTION DASHBOARD

The first pilot was an AI-enabled bridge inspection dashboard. It was intended to address the problem of efficiently getting answers to questions about bridge condition from lengthy (300+ page) and complex inspection reports with hundreds of images. NJDOT worked with Johns Hopkins University (JHU) and Rutgers to create the dashboard with three functions: (1) general question answering (chatbot functionality), (2) detailed information retrieval, and (3) cross-report information prediction (that predicts the lifecycle of a component based on image, text, and numerical information). The tool uses knowledge graphs to knowledge graph to connect key report information such as damages, locations, repair history and improve retrieval accuracy for complex documents.



NJDOT's AI-enabled bridge inspection dashboard, presented by Giri Venkiteela.

NJDOT chose to develop and deploy a locally-hosted, custom tool to support better prediction and deeper analysis, provide a more robust and trustworthy solution, reduce inference costs (compared to a cloud solution), and prevent leakage of sensitive information. They used a mixture-of-experts (MOE) framework, using different AI models (Claude, Gemini, ChatGPT) for different query purposes to provide higher quality and more accurate outputs.

NJDOT found that the tool was able to provide good answers to real questions – such as "What defects were found on Bridge 405 in last 3 inspections?". It reduced report analysis time from hours to minutes, achieved a high level of adoption among inspectors.

AI-ENABLED STRUCTURE DEFECT DETECTION TOOL

The second pilot involved creating an automated inspection tool for detecting and quantifying corrosion defects from images using natural language inputs. Images can be captured by drones or during manual inspections. The system was built with a hybrid framework that integrates vision language models (VLMs) with domain-specific convolutional neural networks (CNNs) to automate detection and evaluation of bridge element defects. A web-based interface was created, supporting queries such as: "what parts of this girder have poor-level corrosion", or "calculate the area distribution of each corrosion condition state".

Initial applications focused on bridge girders and bearings, with expansion planned to concrete and other structural components. One goal is to move from vague descriptions of conditions such as corrosion toward more precise assessments that can support costly decisions such as repainting or replacement.

Venkiteela reported that the system was producing results close to the manually established ground truth, while acknowledging that development and improvement are continuing.

Grace Sammartino — Senior Account Executive, Esri

From Data to Digital Twins: The Future of Asset Management

Sammartino framed her presentation around how transportation agencies can **v**isualize and analyze data products and use them to support the future of asset management.

MOVING INFORMATION ACROSS THE FACILITY LIFECYCLE

She stated that a successful digital environment should allow information to move through the infrastructure lifecycle: survey → design → construction → planning → operations → asset management. She highlighted UDOT as an agency that has made substantial progress toward this type of seamless digital information flow, but stressed that a DOT does not need a perfect 3D digital environment before it can begin realizing benefits from digital delivery.

One recurring challenge is the construction to asset management gap – a disconnect between what was designed, what was actually built, and the inventories used for asset management. Sammartino argued that getting this information flow right is essential for meaningful digital twins and lifecycle asset management.

DIGITAL TWINS

Grace acknowledged that there are differing definitions of "digital twins" and described Esri's view broadly as a spatial-data infrastructure that models and merges the built and natural worlds. She also emphasized that digital twins can create shared understanding and communication, not merely technical simulation. She described digital twins as existing along a maturity continuum .Agencies may begin with basic digital representations of facilities, and then progress towards visualization, simulation, predictive analysis, real time sensing, and increasingly automated responses. She noted that many DOTs are still in the earlier portions of this maturity curve, but can still derive value. She then illustrated a range of possible applications of digital twins:

- **Environmental and Resilience** – combining climate, ecological, and infrastructure information to understand conditions and simulate potential impacts before disasters occur.
- **Planning, Design, and Construction** – visualizing proposed projects, connecting BIM, CAD, GIS, imagery, documents, and field data in a common platform to be shared across disciplines.
- **ITS** – monitoring the operational health of ITS infrastructure and control boxes, including real-time information about system status and latency.
- **Wildlife Collision Prevention** – placing GPS collars on wildlife to track their locations and alerting wildlife managers when they approach a geofence (enabling pre-emptive action to avoid collisions).
- **Airport Management** – integration of CAD, GIS, BIM and imagery at San Francisco International Airport, an application that evolved from descriptive visualization toward real-time information and predictive analytics for airport risk.
- **Multimodal Intersection Analysis** – example from Raleigh, NC of a dynamic, real time operational environment for multimodal intersection analysis, developed through a partnership between Nvidia, Microsoft and Esri. Uses data on vehicle, pedestrian, and other movements (e.g., scooters) to create alerts about potentially hazardous situations that could lead to a collision.

Discussion and Q&A

WHAT ACTUALLY QUALIFIES AS A DIGITAL TWIN?

There was some debate among participants about whether a digital twin must have a3D representation, and whether it must incorporate real-time data.

One participant introduced a National Academies definition emphasizing that a digital twin:

- “mimics the structure, context, and behavior of a natural, engineered, or social system; is dynamically updated from its physical counterpart; has predictive capability; informs decisions that create value; and involves bidirectional interaction between the virtual and physical systems.“

HOW DO AGENCIES GET EMPLOYEES TO TRUST AND USE THESE TOOLS?

Sheehan redirected the discussion from definitions toward implementation.

He asked how agencies integrate difficult and disparate data, develop workforce confidence, train employees, establish trust in AI outputs, and overcome implementation obstacles.

Venkiteela emphasized involving subject-matter experts from the beginning. Because they understand how the system was developed and have shaped its behavior, they are more willing to trust and use it. This was contrasted with introducing employees to a finished “black box,” which can create skepticism because users do not understand what is happening inside the system.

Major observed that staff capability exists on a spectrum. Some employees can move toward more data-science-oriented work, while others can productively use user-interface-based AI and GIS tools. The implication is that not everyone needs to become a data scientist, but transportation professionals increasingly need enough data literacy to work effectively with these technologies.

WHAT ORGANIZATIONAL BARRIER MATTERS MOST?

Sammartino identified executive sponsorship and leadership as the number-one impediment she sees when digital-delivery and AI initiatives fail at DOTs.

This complemented NJDOT’s emphasis on demonstrating ROI: technical teams need both evidence of value and leadership willing to support implementation.

DAY 2 · SESSION 05

Data Fusion: Building the Data Foundation for AI-Enabled Transportation Decisions

Technical Session 6

MODERATOR **Timothy Henry, Professor and AI Program Director, Rhode Island College**

SESSION DESCRIPTION

This session focused on data fusion and the data foundations needed to support analytics, AI, and transportation asset-management decision making. Henry framed data fusion as an increasingly important challenge as agencies move from periodically combining static datasets toward continuously integrating real-time information. He also described the Rhode Island AI Hub’s broader mission of preparing the state’s workforce for an AI-enabled economy.

The three presentations approached the issue from complementary perspectives:

- **Jennifer Volkening of UDOT** described Utah's enterprise data modernization, governance, and architecture for making agency data "AI ready."
- **Justin O'Daniel of Databricks** discussed how agencies can create a governed, reusable data foundation that connects disparate data sources and enables self-service analytics and AI.
- **Justin Bruner of PennDOT** demonstrated how PennDOT is using machine learning and integrated asset-management tools to convert large amounts of bridge, pavement, safety, mobility, and other information into actionable project recommendations.

A common message emerged across all three presentations: DOTs generally do not lack data. Their challenge is connecting, governing, understanding, and translating that data into decisions.

SESSION TAKEAWAYS

- **Data fusion is not primarily about putting all of an agency's data into one enormous database.** It is about creating the architecture, standards, governance, lineage, and organizational relationships that allow information from different sources to be combined when needed and trusted when used.
- **AI readiness starts with data readiness.** UDOT is deliberately modernizing its legacy systems and strengthening data governance before attempting to scale AI applications. Its vision is "reliable data that empowers every decision."
- **Data should be managed as an organizational asset.** UDOT's principles include clear authority and accountability, proactive data-quality management, governance embedded in operations, risk- and value-based management, and security and privacy by design.
- **The fundamental problem is a "data-to-decision gap," not a data shortage.** DOT information often resides in numerous systems, formats, organizational units, and refresh cycles. Agencies may have enormous quantities of data while still being unable to see the whole system, trust the numbers, prioritize investments, anticipate problems, or efficiently generate insights.
- **A shared architecture can preserve specialized systems while connecting their information.** Both UDOT and Databricks described medallion architectures that progressively transform raw data into increasingly refined, governed, and decision-ready information.
- **Modernization does not necessarily mean replacing everything.** O'Daniel advocated keeping systems that provide value while reducing fragile integrations, duplicate data, and repeated manual preparation.
- **Start with a high-value use case rather than a "big bang" modernization.** Once the data required for one application has been connected, cleaned, governed, and documented, that foundation can be reused for subsequent applications.
- **Natural-language analytics could dramatically broaden access to agency data.** Databricks' Genie example allows authorized users to ask questions of governed data in plain English rather than waiting for specialized staff to generate a report or data extract.
- **Human review and AI guardrails remain essential.** PennDOT's system proposes and bundles treatments, but users can inspect, modify, validate, and approve them.
- **Traceability builds trust.** PennDOT records the decisions made by its automated processes so staff can determine why a recommendation was generated and diagnose errors.
- **Open-source collaboration can reduce duplication across DOTs.** PennDOT makes its Asset Fox platform available through GitHub and a pooled-fund effort so other agencies can build on work already performed rather than recreating it independently.

Tim Henry — Professor and AI Program Director, Rhode Island College / Academic Team Lead for Rhode Island Governor's AI Task Force

Tim Henry opened by describing the mission of Rhode Island's AI Hub, whose primary charge is workforce development.

The rapid emergence of generative AI has created an immediate need to train large numbers of people. The Hub is already working with K–12 and higher-education faculty, nonprofits, small businesses, larger corporations; and current members of the workforce. It is also working with the education system so that future graduates enter the labor market prepared for an AI-enabled workplace.

A second priority is improving AI literacy among Rhode Islanders by providing accessible information about what AI is and where people can obtain free training and resources.

A third priority is facilitating partnerships among government, higher education, and the private sector.

Henry noted that Rhode Island's Governor's AI Task Force identified an AI-ready workforce as a potential differentiator for the state. He connected this workforce mission to the session topic through his own experience with an earlier static data-fusion project. Today's challenge is substantially more complex because applications such as digital twins may require continuous, real-time data fusion rather than periodically combining static datasets.

Jennifer Volkening — Director of Data Analytics and Governance, UDOT

From Governed Data to Responsible AI

Volkening, UDOT's opened with UDOT's vision for providing stakeholders with "access to reliable and accurate data to enable and support data-driven decision-making, performance management, legal and regulatory compliance requirements, while achieving transparency". To achieve this, UDOT treats data as an organizational asset and emphasizes:

- defined authority and accountability;
- proactive management of data quality;
- integration of governance into operations and organizational culture;
- management based on risk and value; and
- security and privacy by design.

UDOT is undertaking a major modernization effort involving a legacy system originally developed in-house more than 40 years ago. Over time, numerous other systems became connected to it, resulting in fragmentation of information and complex integrations. UDOT decided approximately three years ago to replace the legacy environment with a unified business platform structured as a system of systems. The existing system is being replaced by three systems using a modular microservices architecture. These systems can communicate bidirectionally while relevant data flows into UDOT's Advanced Data Analytics Platform, or ADAP, built around BigQuery.

Before selecting replacement systems, UDOT assessed its data-governance maturity. The agency determined that it was between maturity Levels 2 and 3. Its immediate ambition is to reach approximately Level 4 through the modernization initiative.

UDOT's modernization effort is using a balanced framework that defines the technical design (medallion architecture), the organizational responsibilities (data mesh), and the governance. The medallion architecture provides a design pattern to organize data within a storage system, focusing on cleanliness and quality. The data mesh establishes domain ownership, and provides a data self-serve infrastructure. The governance is an operational model providing automated guardrails to prevent a "data mess". The intent is to decentralize governance while maintaining checks and balances among business owners, business-area data analysts, and IT staff responsible for implementation.

UDOT adapted the traditional three-level medallion architecture into four levels.

- **Bronze** contains raw source data essentially as received.
- **Silver** is where much of the work occurs. Data is cleaned, joined and merged; metadata and lineage are established; and relationships among disparate datasets are developed.
- **Gold** contains highly refined, trusted data that can be made broadly accessible within UDOT.
- **Platinum**, an additional UDOT layer, contains highly curated data configured for specific functional applications.

As data progresses through these levels, its quality, refinement, aggregation, and trustworthiness increase.

UDOT is also establishing data lineage through its GitHub environment and using tools including Apache Airflow/Cloud Composer to automate movement from source systems through extraction, computation, ingestion, storage, and ultimately BigQuery.

UDOT's modernization is explicitly connected to its publicly available AI strategy and three-phase roadmap. The three phases are:

- **Foundation:** involving agency alignment and discovery, strengthening the data infrastructure and governance, building foundational AI literacy, training staff on AI tools, and developing early cost-saving and efficiency-focused AI products.
- **Growth and Integration:** involving expanding partnerships, integrating AI into transportation system improvements, and advancing predictive and user-centered AI use cases.
- **Scale and Optimize:** involving scaling successful AI applications across the department; continuous monitoring and improvements to AI systems; and enhancing safety, operations, and customer experience through mature AI solutions.

The agency is currently implementing the initial phase of that roadmap and had recently completed an AI Discovery Tour.

Justin O'Daniel — Enterprise Account Executive, Databricks

Closing the Data-to-Decision Gap

Justin O'Daniel began by arguing that DOTs do not have a shortage of data, analytics tools, GIS applications, automation, or AI ideas. They have a data-to-decision gap.

Transportation executives commonly encounter problems such as:

- being unable to see the entire transportation system in one place;
- not fully trusting the data behind decisions;
- difficulty prioritizing constrained funding;
- discovering problems too late; and

- devoting significant staff capacity to reporting without generating corresponding insight.

These are not simply IT problems. They affect safety, mobility, cost, investment decisions, and public accountability.

Mission-critical information exists across many systems, departments, formats, owners, and update cycles. Over time this has produced duplicative information, multiple versions of the truth, inconsistent definitions, limited lineage, and governance rules residing in documents rather than being embedded in the systems themselves.

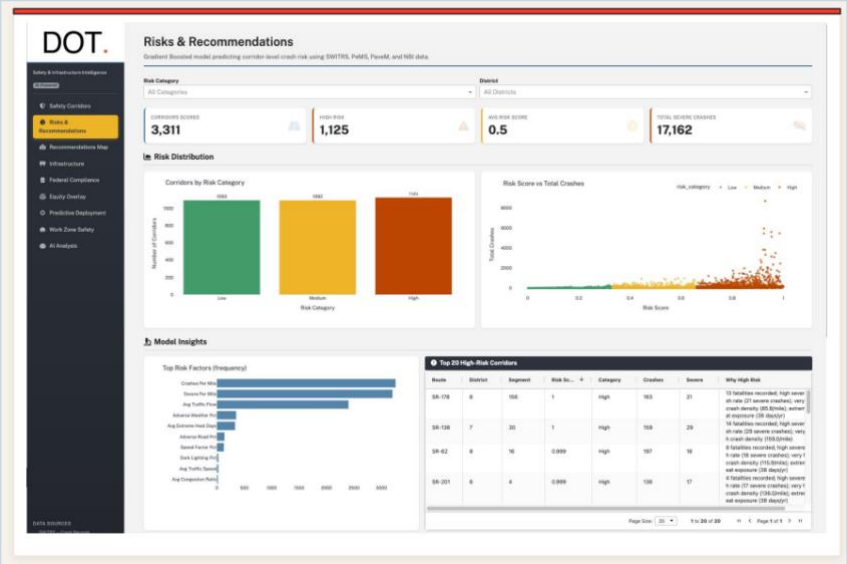
O'Daniel stressed that approaches such as data warehouses, data lakes, departmental data marts, point-to-point integrations, and standalone AI pilots have created local value, but frequently **stall at scale** because each new initiative recreates much of the underlying work.

Databricks' proposed alternative is a governed "lakehouse" architecture combining characteristics of data warehouses and data lakes. O'Daniel described the familiar medallion progression:

- **Bronze:** preserve raw data and its history.
- **Silver:** clean, connect and standardize data; establish quality rules, lineage, and access controls.
- **Gold:** produce decision-ready data products organized around domains such as safety, asset condition, traffic operations, work zones, capital programs, and resilience.

Rather than creating new standalone interfaces, governed information can be delivered back into the maps, dashboards, federal reports, traffic-management systems, and other applications employees already use.

O'Daniel demonstrated a conceptual "Transportation 360" environment integrating crash, traffic, pavement, bridge, climate, and real-time work zone data within a GIS environment. Such a foundation could support applications including safety, infrastructure management, federal compliance, predictive maintenance, capital planning, resilience, and AI analysis.



A "Transportation 360" environment linking crash, asset, traffic, and work-zone data, presented by Justin O'Daniel.

He provided an example that used multiple datasets to identify a high-risk corridor, explain the evidence behind the risk score, recommend potential countermeasures, and connect those countermeasures to potential federal funding programs.

He stressed that the goal is not for the model to make the engineering decision. It is to give engineers and grant teams **a** traceable, evidence-based starting point.

O'Daniel advised agencies not to begin with the most ambitious use case. He suggested self-service analytics as a practical entry point.

He explained how Databricks' Genie allows authorized users to ask plain-language questions of governed data. Instead of submitting a data request and waiting days or weeks, a manager could ask questions in plain-language – such as "Which corridors have the highest severe-crash rate per mile?" The important feature, he stressed, is not that Genie is another chatbot. Its value comes from being grounded in approved definitions, governed data, and access controls. He cited an example in which a transportation organization reduced a roughly two-week analytics turnaround to seconds.

Justin Bruner — Asset Management Engineer, PennDOT

Using AI to Move from Asset Data to Projects

Bruner began his presentation by challenging the common goal of simply "breaking down silos." Instead, he described specialized data systems as "cylinders of excellence." He argued that Bridge specialists need bridge information structured appropriately for bridges; pavement specialists need pavement information suited to their work. The objective should not necessarily be to eliminate those domain-specific environments, but to connect them to can support broader decisions.

Bruner's presentation described PennDOT's development of their three interconnected modules for asset management: Asset Fox, Project Builder, and Decision Space.

ASSET FOX

Asset Fox is an enterprise asset management platform that consumes asset information, applies machine learning, and produces deterioration forecasts, treatment recommendations, and financial projections. A particularly useful output is a simple forecast showing historical and projected good/fair/poor condition under different investment levels. Bruner noted that this is highly effective for communicating with executives because it connects funding decisions directly with future asset condition.

Asset Fox is not an inventory management system – PennDOT has two home-grown systems for this: PAMS for pavement and BAMS for bridges.

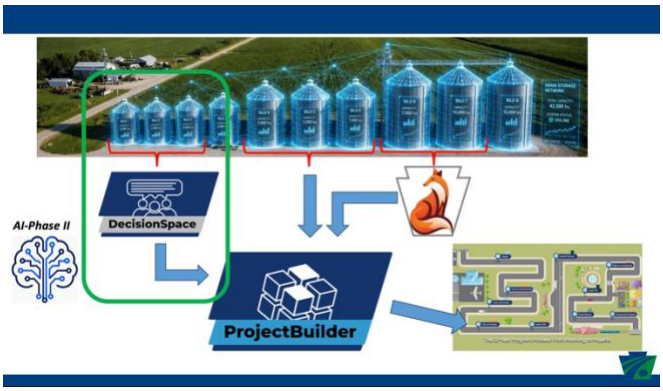
PROJECT BUILDER

Project Builder is a separate application that consumes recommended bridge and pavement treatments from Asset Fox and automatically assembles them into candidate projects. These candidates are then reviewed and adjusted (as needed) by PennDOT staff. For example, a user might discover that the system assigned an unrealistically low cost to a bridge preservation treatment. The planner can edit the treatment or cost, save the change, and have the revised information flow into the project-development process.

DECISION SPACE

PennDOT created Decision Space to bring candidate projects that target capacity, safety, accessibility, or other types of needs into the same planning environment. Decision Space compiles information on project characteristics, costs, and impacts. The system can then score, compare, and prioritize candidates while still allowing human review.

Bruner sees significant future potential for a localized small language model to search agency information, populate candidate-project information, and help staff evaluate alternatives within carefully established guardrails.



AI-Integrated Decision-Making Presented by Justin Bruner.

OPEN SOURCE AND POOLED-FUND COLLABORATION

PennDOT makes Asset Fox available free through an open-source GitHub repository and has organized a pooled-fund effort involving other transportation agencies.

Bruner argued that DOTs historically have developed useful tools independently and then struggled to share them. Open-source development gives agencies an opportunity to save each other time and money rather than repeatedly solving the same problems.

Discussion and Q&A

WHY MAKE PENNDOT'S TOOLS OPEN SOURCE?

Bruner said the rationale is fundamentally about interagency efficiency. DOTs have traditionally developed tools inside organizational silos even when other agencies face nearly identical problems.

Sharing successful software allows agencies to avoid duplicated development, learn from one another, improve tools collectively, and use public dollars more efficiently. He saw essentially no downside to sharing the work PennDOT has already developed.

WHAT TYPES OF DATA ARE HARDEST TO INTEGRATE?

O'Daniel identified unstructured data as one of the largest gaps. Traditional enterprise data warehouses primarily handle structured data, while agencies increasingly need to incorporate documents, imagery, text, and other unstructured information. Doing so through separate tools can create still more systems and processes. His preferred approach is to incorporate both structured and unstructured information into the same governed environment.

Volkening provided a UDOT example involving crash data. Previously, the process relied on file-based extraction followed by manual spreadsheet cleanup. UDOT now has a native connection to the university providing the data and can bring it into a restricted environment, clean it, protect PII and sensitive information, and then move standardized data through the medallion architecture. The result is a more consistent and repeatable crash-data pipeline.

HOW DO AGENCIES HANDLE EXTERNAL, VENDOR, OR PRIVATE DATA?

O'Daniel described Databricks capabilities for sharing data with external organizations while preserving controls. Sensitive information can be masked, and "clean rooms" can allow another party to perform permitted analytics without receiving or owning a copy of the underlying data.

Bruner described a deliberately simpler PennDOT approach: Excel in, Excel out. PennDOT's platform is not directly connected to external systems. Information is imported and exported through Excel, avoiding potentially expensive commercial integrations and ongoing maintenance dependencies.

HAS PENNDOT'S AUTOMATED PROJECT DEVELOPMENT WORKED IN PRACTICE?

Bruner said the underlying Asset Fox platform has been operating at PennDOT for approximately seven or eight years, making the bridge component mature and trusted. The pavement implementation is newer and still requires refinement of treatments and consequences. Project Builder itself was extremely new at the time of the symposium—roughly ten days old—but initial stress testing suggested that it was grouping treatments logically, remaining within cost constraints, respecting geographic constraints, and carrying appropriate treatments from Asset Fox into candidate projects. PennDOT was systematically testing dozens of variables and limits.

Bruner attributed much of the system's success to traceability. Every automated decision is recorded so developers can inspect the logic and determine precisely where a process worked or failed.

DAY 2 · SESSION 06

Conference Takeaways and the Path Forward for AI in Transportation

Peer Roundtable 2

SESSION DESCRIPTION

The final session of the RIDOT AI Symposium was designed as a reflection and synthesis of the two-day event rather than a set of formal presentations. A cross-section of transportation leaders from state DOTs, USDOT, AASHTO, TRB, and other organizations were asked to identify their most important takeaways, after which the discussion was opened to the audience.

The conversation moved from reflections on the symposium to a substantive discussion of what the transportation community should do next. Major themes included focusing on real problems rather than AI hype; accelerating implementation; addressing organizational change and workforce needs; understanding the full cost of AI; sharing lessons and resources more effectively; expanding peer exchange internationally; and creating a coordinated transportation AI knowledge network.

The session ultimately reinforced one of the strongest themes of the symposium: the central challenge is increasingly not whether AI technology can do useful things, but whether transportation organizations can adopt, govern, afford, sustain, and effectively use it.

SESSION TAKEAWAYS

- **Start with the transportation problem, not the technology.** Multiple speakers returned to the question: *What problem are we trying to solve, and can AI help us produce a better outcome?* The pace of technological change makes four-year implementation cycles untenable; agencies should focus on what can be accomplished now or within the next six months.
- **Be skeptical of AI hype while learning aggressively from actual applications.** Transportation has repeatedly encountered purported transformational technologies—from Transportation Network Companies (TNCs) to automated vehicles. We should not assume that AI will automatically transform or "save" transportation. The more useful question is what agencies are learning from applications already underway.
- **The organizational adoption curve is much slower than the technology curve.** AI is advancing exponentially while public-sector organizations change much more slowly. Closing that gap may be one of the transportation sector's most important challenges.
- **The human side may be harder than the technology.** " AI needs to not only have the "human in the loop" but it needs to be "human centered and human focused." One panelist characterized the challenge as roughly "20% technology and 80% human. Agencies need to address trust, ethics, resistance to change, employee comfort, training, and the perception that using AI somehow constitutes "cheating."
- **AI should improve public service, not simply automate work.** The appropriate test is whether AI helps agencies produce better projects, decisions, services, safety, or outcomes—not whether it replaces tasks humans currently perform.
- **AI has operating and maintenance costs that agencies need to budget for.** A panelist highlighted token costs as an emerging issue, citing organizations receiving unexpectedly large bills. He compared token consumption to an infrastructure asset's O&M costs—something that must be understood before deployment rather than discovered afterward.
- **Transportation needs a better mechanism for sharing AI knowledge.** There are already numerous pilots, repositories, research efforts, and demonstrations, but no easy way for practitioners to understand what exists, what worked, who is doing similar work, and whom they should contact.
- **The answer should be a knowledge network, not simply a document repository.** USDOT participants described a planned AI "portal" that would use LLM-assisted discovery, community engagement, links to open-source resources, and connections among practitioners rather than simply becoming a digital bookshelf of PDFs.
- **USDOT, AASHTO, TRB, professional associations, states, universities, and industry all have complementary roles.** The emerging concept was deliberately federated and decentralized rather than controlled by one organization.
- **Peer exchange needs to extend beyond U.S. state DOTs.** A panelist emphasized that significant AI innovation is occurring internationally and cited examples discussed during the symposium from China, Singapore, Austria, and many other countries.
- **AI workforce development should focus on competencies, not simply new job titles.** Transportation agencies do not need every employee to become a data scientist. They need to identify what level of AI and data competency each type of employee requires.
- **Existing employees may already possess many of the needed skills.** GIS professionals were highlighted as an example: they understand data models, schemas, business requirements, agency processes, and multiple business units, making them natural candidates for upskilling into more AI- and data-intensive roles.
- **AI could democratize analytical work—or inadvertently narrow the workforce.** A panelist cautioned against redefining transportation jobs so narrowly around computer science and technical credentials that agencies reverse progress toward a broader and more inclusive workforce.

- **AI should eventually become an ordinary tool.** One speaker compared AI to Excel, Word, and the internet—technologies once limited to particular specialists that eventually became ubiquitous workplace tools.
- **Do not "boil the ocean."** The closing audience comments reinforced the need to focus AI investments on specific, achievable problems and build from demonstrated successes.
- **Continued collaboration is needed.** Panelists agreed on the need for continuing conversations and that multiple agencies can play a role: USDOT can provide national coordination; AASHTO can reach state DOT practitioners through its committees and communities; TRB can connect government, academia, and the private sector; other professional associations can serve their own specialized communities. Participants also suggested more informal communication vehicles—such as a recurring podcast or short-form updates featuring practitioners discussing what their agencies are actually doing—rather than relying exclusively on long formal reports.